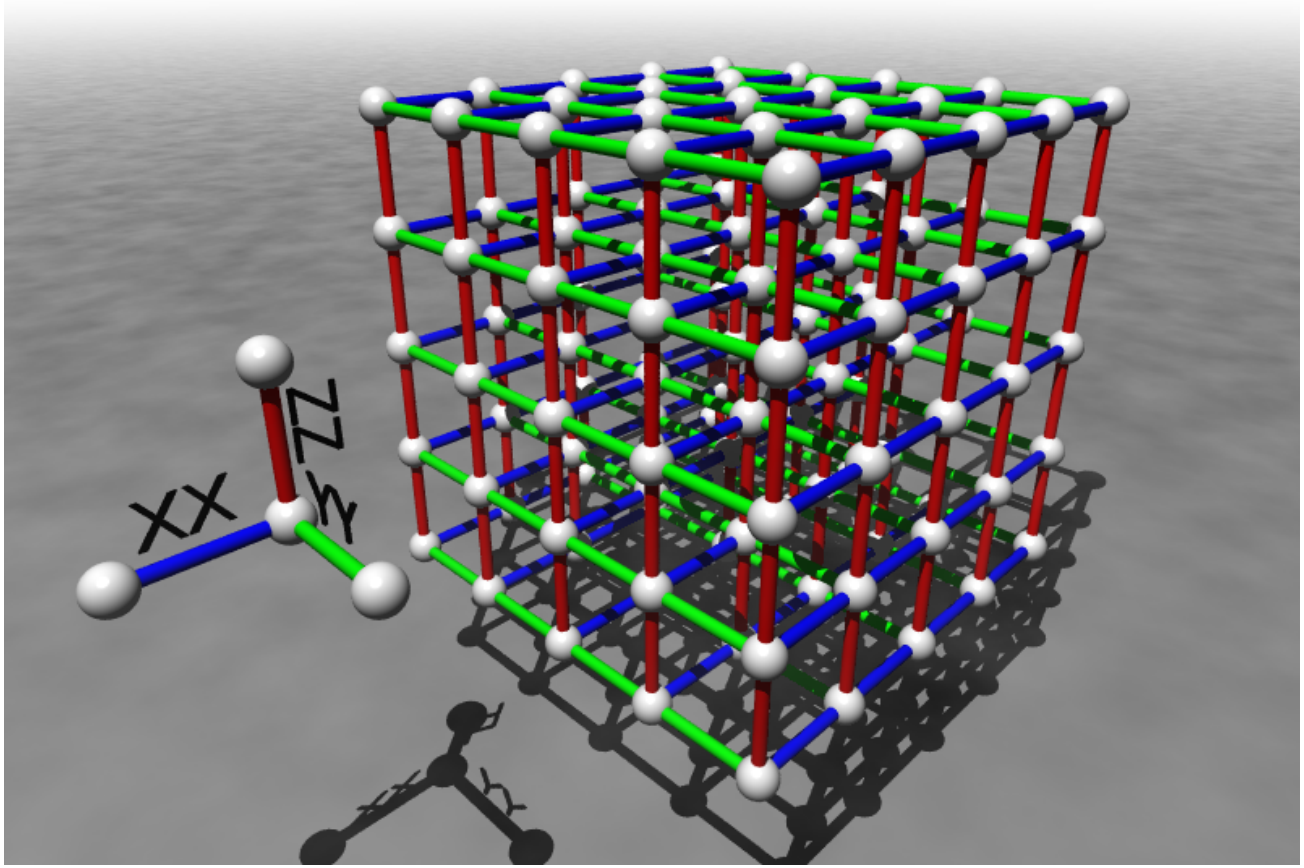
Building Robust Qubits Using Many-Body Strongly Interacting Quantum Systems



Dave Bacon, Principal Research Scientist
Department of Computer Science & Engineering
University of Washington



Happy Valentine's Day



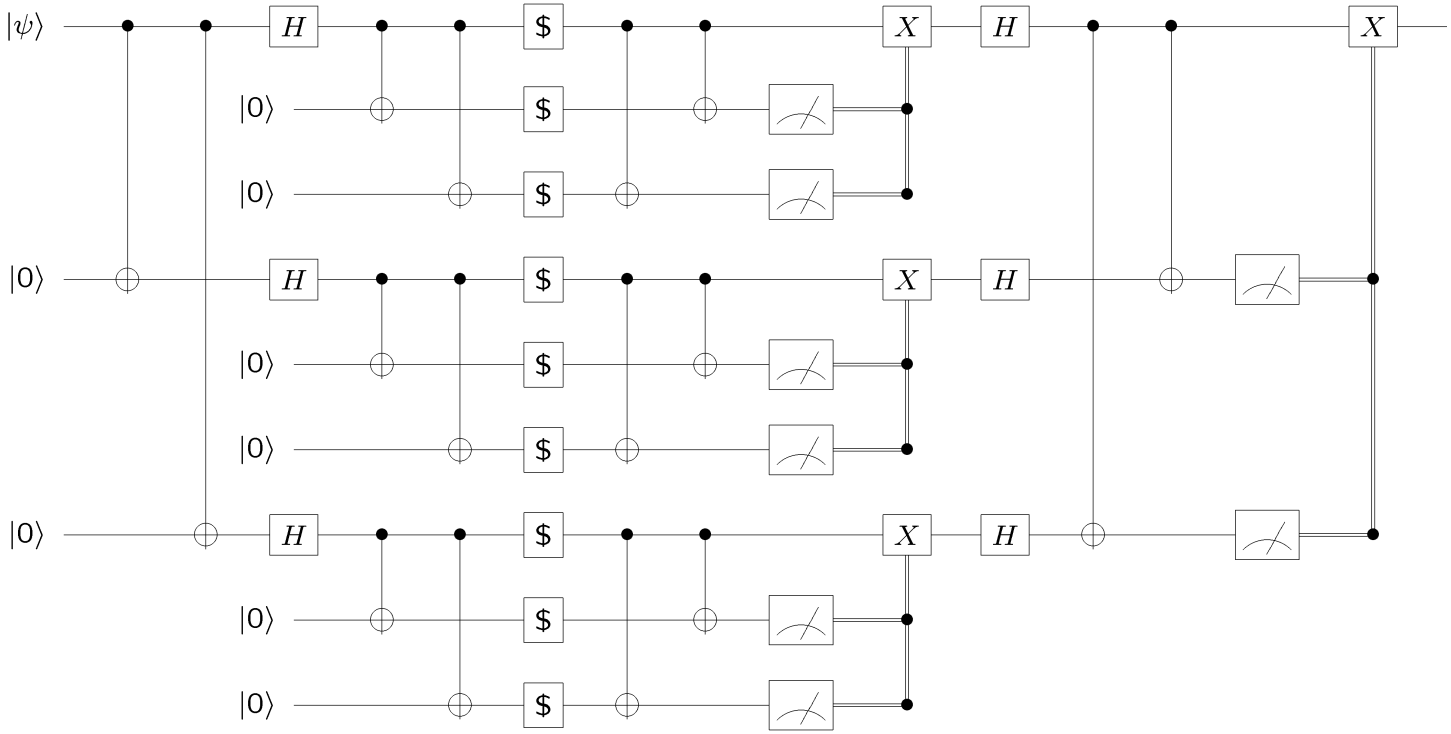
Quantum World, Quantum Computers

“Nature isn't classical, dammit, and if you want to make a simulation of Nature, you'd better make it quantum mechanical, and by golly it's a wonderful problem, because it doesn't look so easy.”

– Richard Feynman 1981



Dirac and Feynman



Shor's Algorithm (1994)



Peter Shor

188198812920607963838697239461650439807163563379417382700763356422988859715234665485319060606504743045317388011303396716199623212057340318795506569622130516875930765025705 9

=

398075086424064937397125500550386491199064362342526708406385189575946388957261768583317

×

472772146107435302523623071973048224632914695302097116459852171130520711256363590397527

FACTORING

Best classical algorithm takes time $O(\exp(n^{1/3}))$

Shor's quantum algorithm takes time $O(n^3 \log n)$

Shor's algorithm breaks the most widely used public key cryptosystem (RSA, Diffie-Hellman)

Perspective

GREATEST COMMON DIVISOR



Euclid's Algorithm

gcd*(1020,153)*

1020 = 6×153+102

gcd*(1020,153)*=gcd*(153,102)*

153 = 1×102+51

gcd*(153,102)*=gcd*(102,51)*

102 = 2×51

gcd*(102,51)*=51



"The problem of distinguishing prime numbers from composites, and of resolving composite numbers into their prime factors, is one of the most important and useful in all of arithmetic... The dignity of science seems to demand that every aid to the solution of such an elegant and celebrated problem be zealously cultivated" — Carl Gauss

Two Quests

What can quantum information processing devices do?



How can we build quantum information processing devices?

Some Notable Recent Results

Efficiently solve Pell's equation [Hallgren 2002]

New efficient methods for density matrix renormalization group in high dimensions [Vidal, Verstraete *et al.* 2003-now]

Quantum Shannon theory [various authors]

Two Divisions

Theory of fault-tolerant quantum computing

"engineering"

Experimental implementations of quantum computers

"physics"

Building Battle Royale



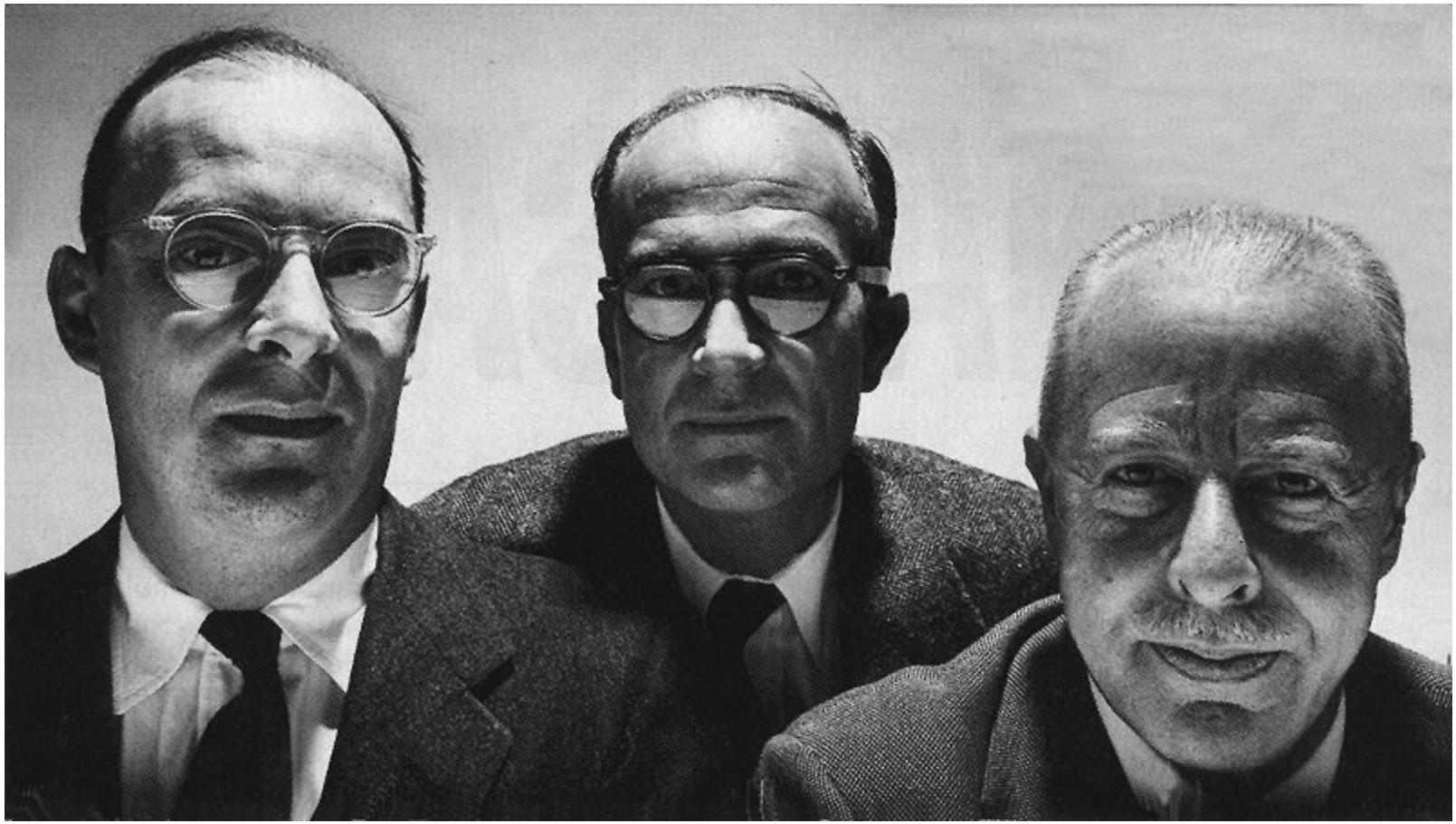
Computer Science & Engineering



Physics

Who will dominate construction of a large scale quantum computer?

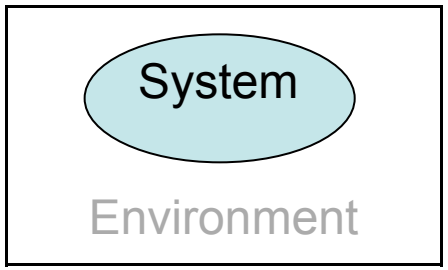
A Different Path



I will argue that the path to constructing a robust large scale quantum computer will look nothing like what is currently envisioned, but instead will be achieved via a careful application of physics to the problems of building robust quantum computers.

Problem Child

decoherence:



$$H = H_S \otimes I + I \otimes H_B + H_I$$

open quantum systems can become entangled with their environment

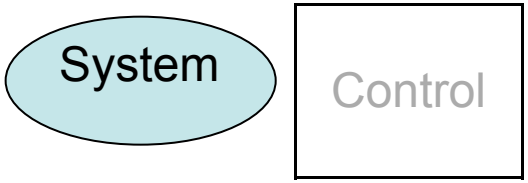
$$(\alpha|0\rangle_S + \beta|1\rangle_S) \otimes |0\rangle_E \rightarrow \alpha|0\rangle_S \otimes |0\rangle_E + \beta|1\rangle_S \otimes |1\rangle_E$$

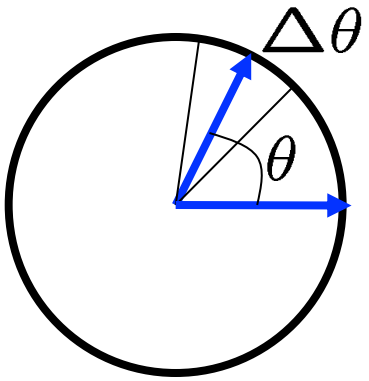
coherent superposition — incoherent sum

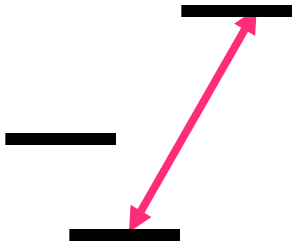
$$(\alpha|0\rangle_S + \beta|1\rangle_S) \rightarrow \rho = |\alpha|^2 |0\rangle_S\langle 0|_S + |\beta|^2 |1\rangle_S\langle 1|_S$$

imprecise and faulty control:

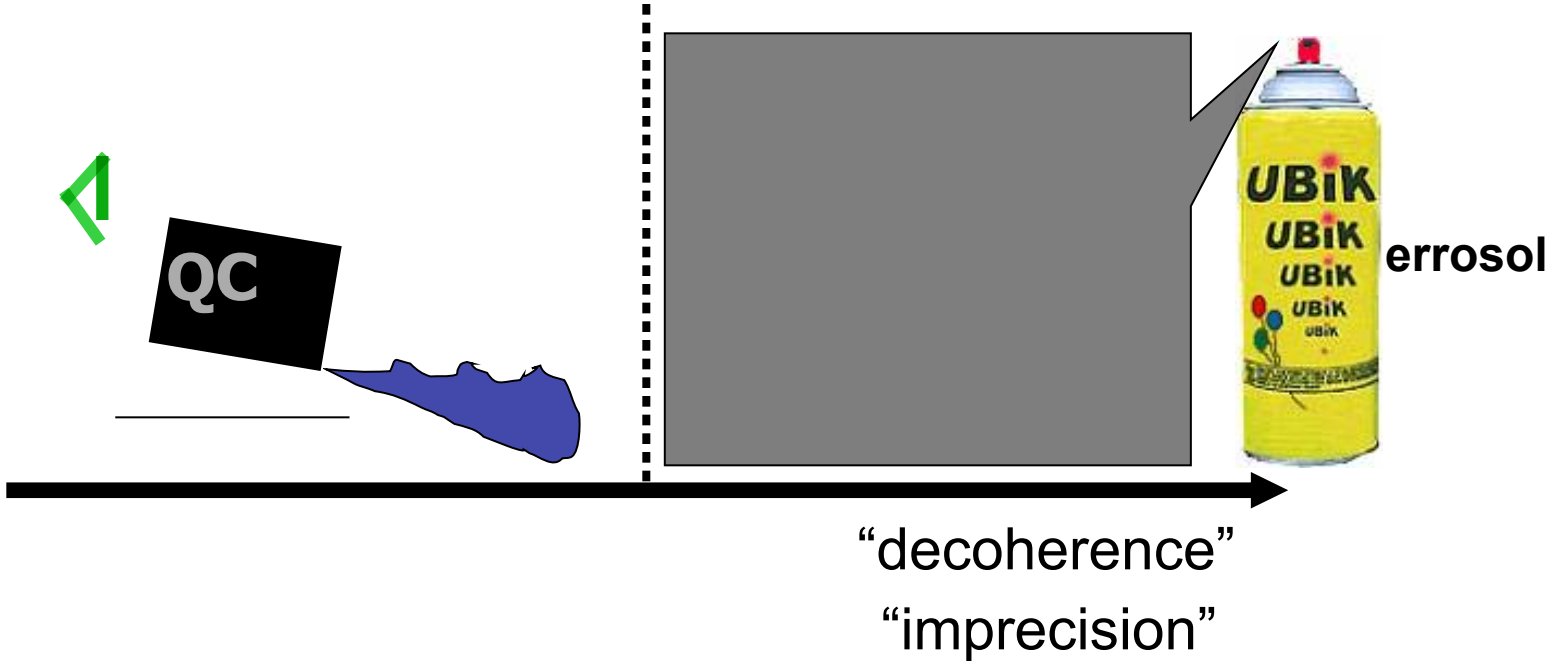
unitary gates, preparation, measurements







Problem Solution

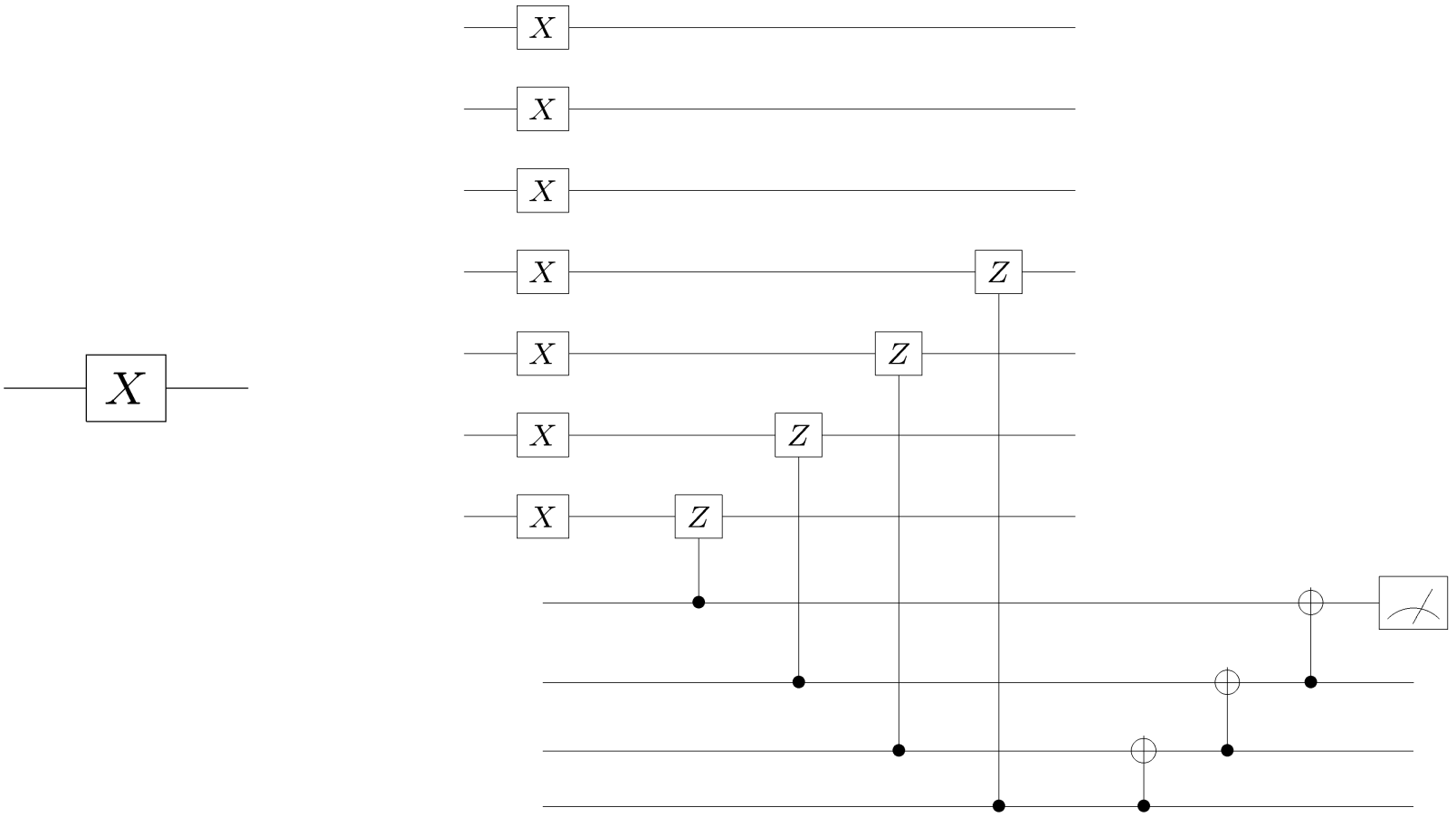


Threshold Theorem for Fault-Tolerant Quantum Computation

[Aharonov, Ben-Or, Knill, LaFlamme, Zurek, Shor, Gottesman, Preskill]

If decoherence is weak enough and control is precise enough, then a quantum computation to a desired accuracy can be carried out with an overhead polynomial in the log of the accuracy.

Sketch: How It Works



To make robust elements, information is encoded into a quantum error correcting codes of significant size and fault-tolerant methods are used for gates, preparation, measurement, and correction

In Practice?

How to build a 300 bit, 1 Gop quantum computer

Andrew M. Steane
Centre for Quantum Computation, Department of Atomic and Laser Physics, Clarendon Laboratory, Parks Road, Oxford, OX1 3PU, England
(Dated: December 22, 2005)

13000 ions, 1000 laser beams, 160000 electrodes in ion trap

The device complexity simply to engineer robust qubits is severe

“A quantum computer will spend most of its time doing error correction.”

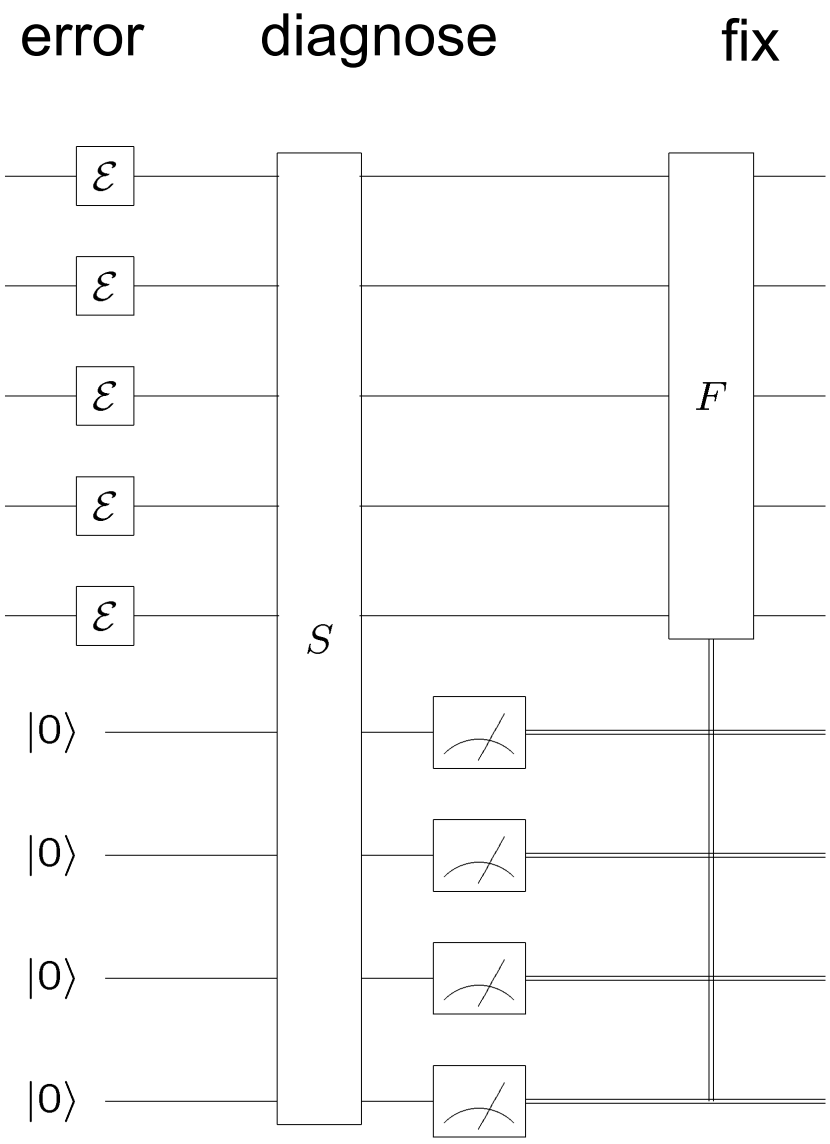
CONTRAST:





Is there a better way? Can we mimic classical computers?

Quantum Error Correction



encoded quantum information

encoded quantum information

"cold" ancilla qubits

"hot" ancilla qubits

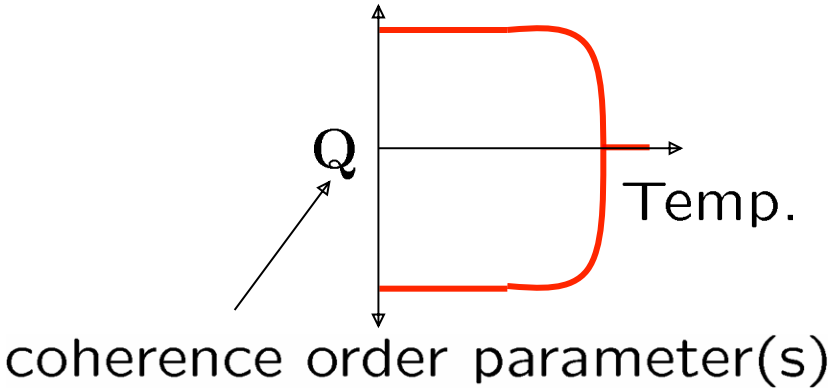
Quantum error correction is essentially (sophisticated) cooling

The Quantum Hard Drive?

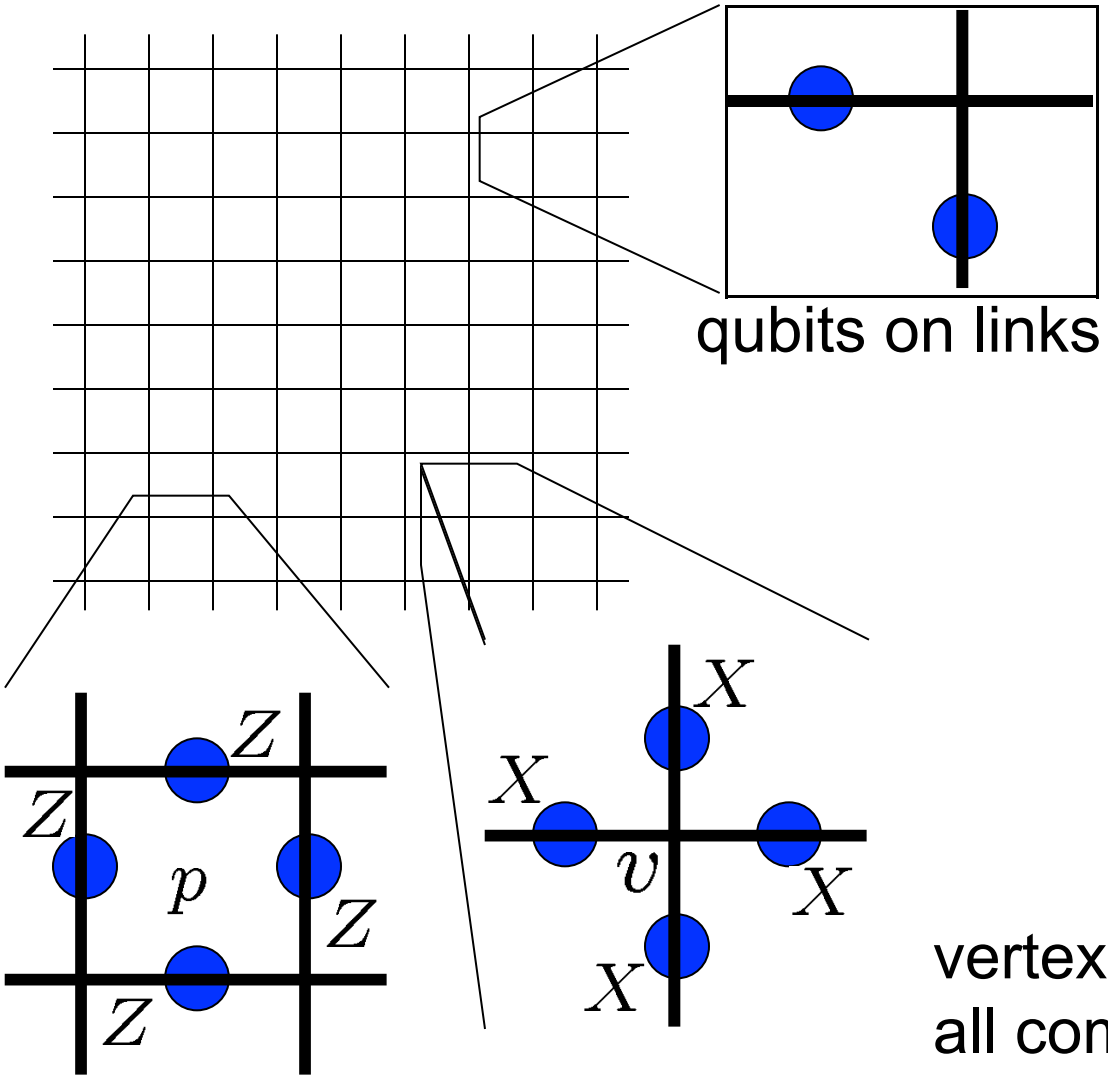


Kitaev

Do there exist (or can we engineer) quantum systems whose physics guarantees fault-tolerant quantum computation?



Kitaev's Model

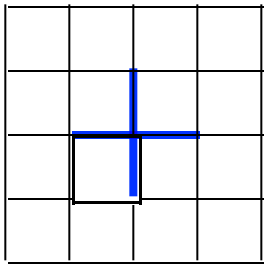


plaque operators:

$$S_p = \prod_{e \in p} Z_e$$

vertex operators:

$$S_v = \prod_{e \in v} X_e$$



vertex and plaque operators all commute with each other

$$Z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \quad X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

Kitaev's Model, Continued

Hamiltonian

$$H = -J\left(\sum_p S_p + \sum_v S_v\right)$$

$$S_p|\psi\rangle = s_p|\psi\rangle \quad s_p \in \{-1, +1\}$$

$$S_v|\psi\rangle = s_v|\psi\rangle \quad s_v \in \{-1, +1\}$$

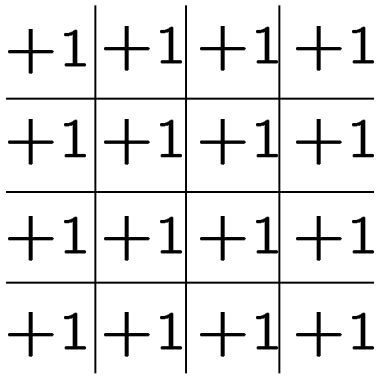
plaque operators:

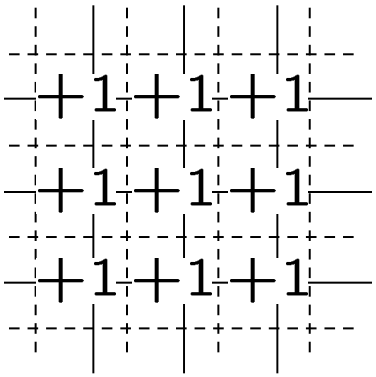
$$S_p = \prod_{e \in p} Z_e$$

vertex operators:

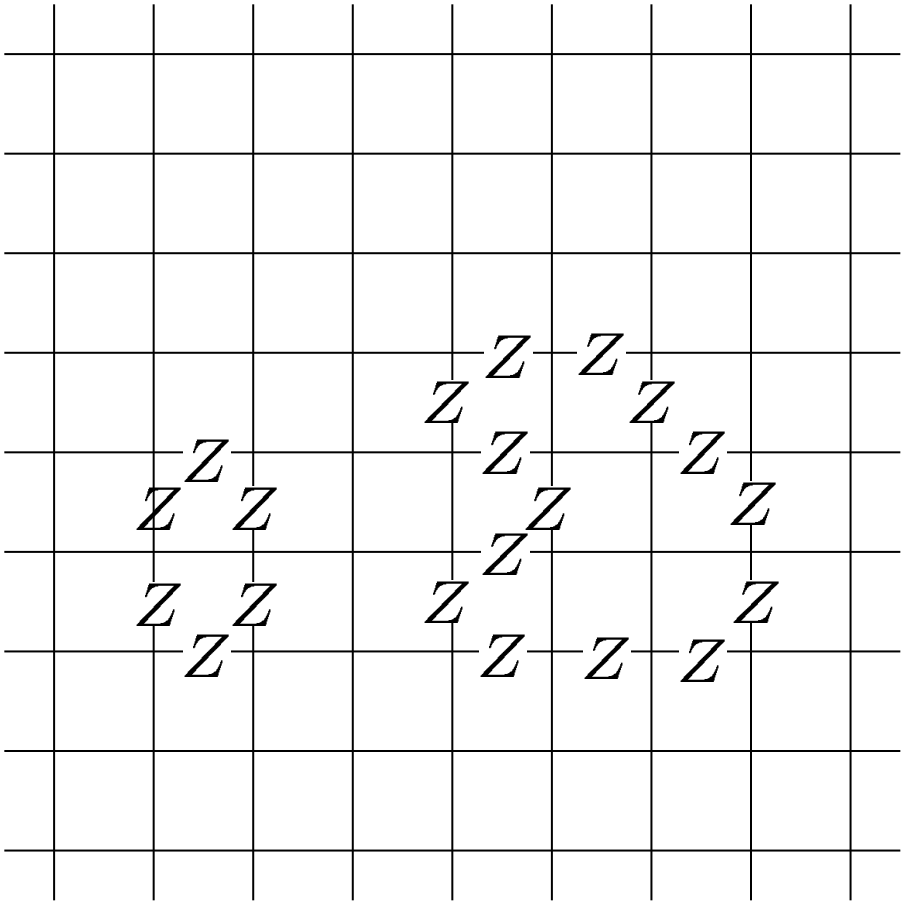
$$S_v = \prod_{e \in v} X_e$$

ground state has all $s_p = s_v = +1$





Topology of Operators


Z Z Z

Degeneracy

Hamiltonian

$$H = -J\left(\sum_p S_p + \sum_v S_v\right)$$

plaque operators:

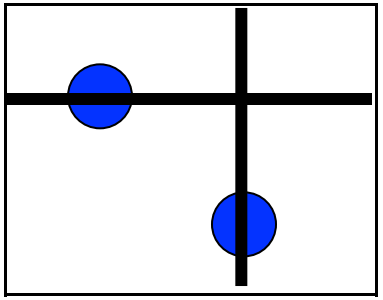
$$S_p = \prod_{e\in p} Z_e$$

vertex operators:

$$S_v = \prod_{e\in v} X_e$$

$2l^2$ qubits

$\dim\mathcal{H} = 2^{2l^2}$

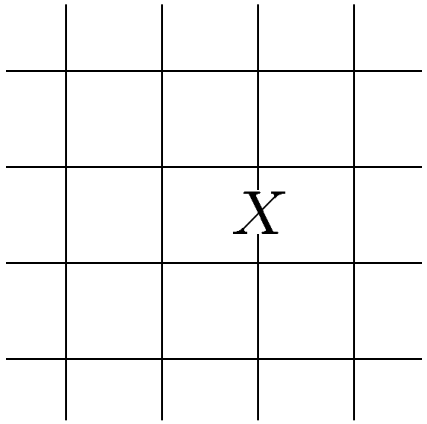


l^2 plaque operators $\prod_p S_p = I$ 2^{l^2-1} $s_p \in \{+1,-1\}$ values

l^2 vertex operators $\prod_v S_v = I$ 2^{l^2-1} $s_v \in \{+1,-1\}$ values

every energy level is $2^{2l^2} - 2^{l^2-1} - 2^{l^2-1} = 4$ degenerate

Kitaev's Model



plaque operators:

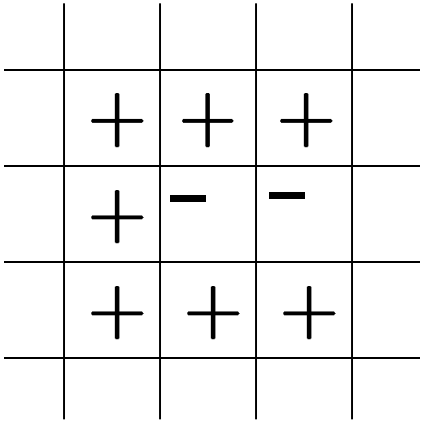
$$S_p = \prod_{e \in p} Z_e$$

vertex operators:

$$S_v = \prod_{e \in v} X_e$$

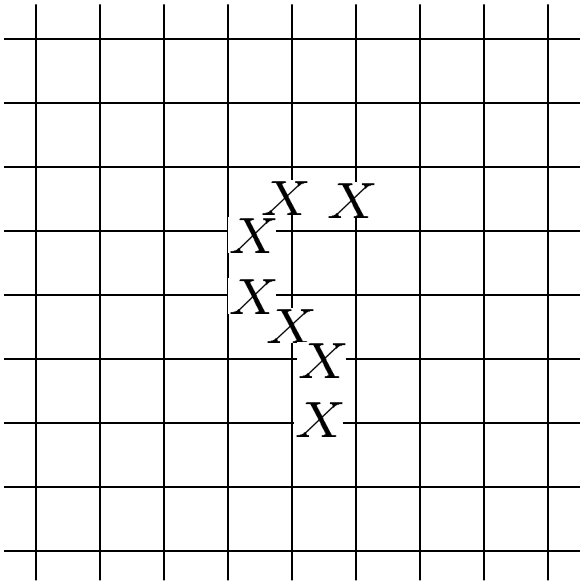
X anti-commutes with two plaque operators

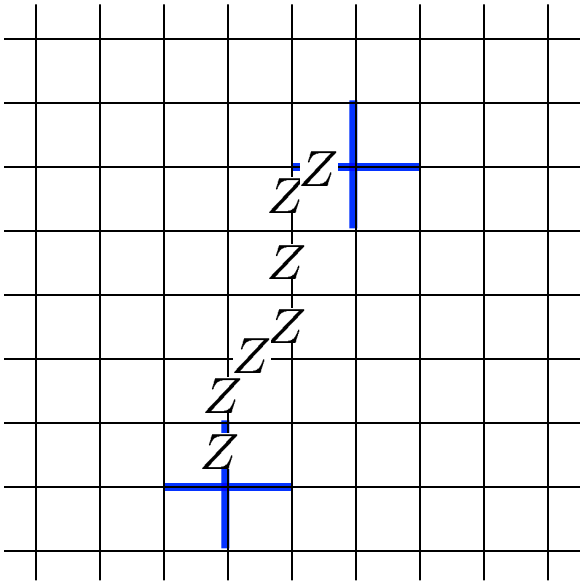
$$\{X, S_p\} = 0 \Rightarrow S_p(X|\psi_G\rangle) = -X|\psi_G\rangle$$



excitation is $4J$ above ground state

Excitations

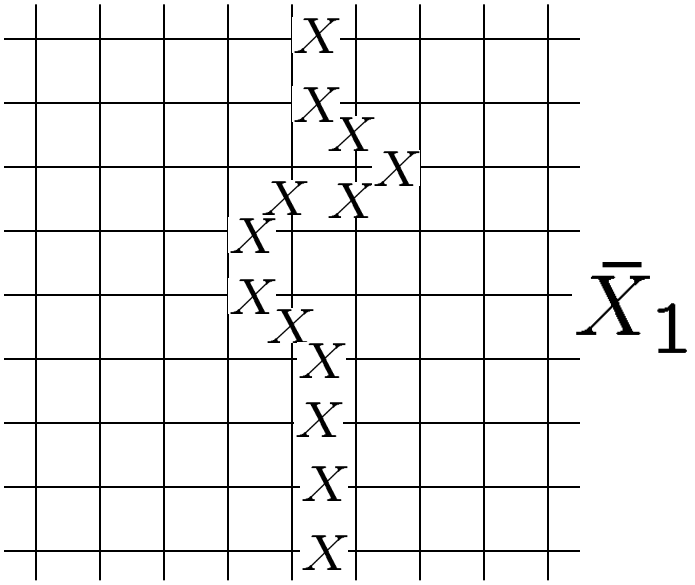


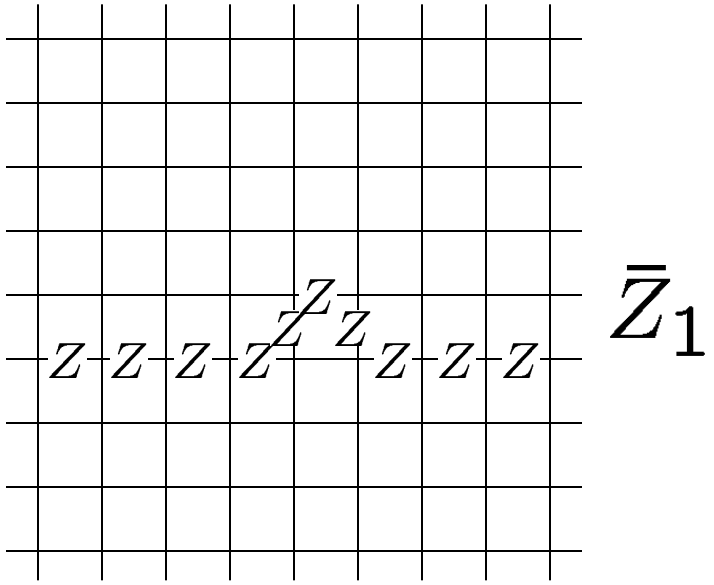


excitations particles come in pairs (particle/antiparticle)
at end of "error" chains

two types of particles, X-type (live on vertices of dual lattice)
Z-type (live on vertices of the lattice)

A Qubit



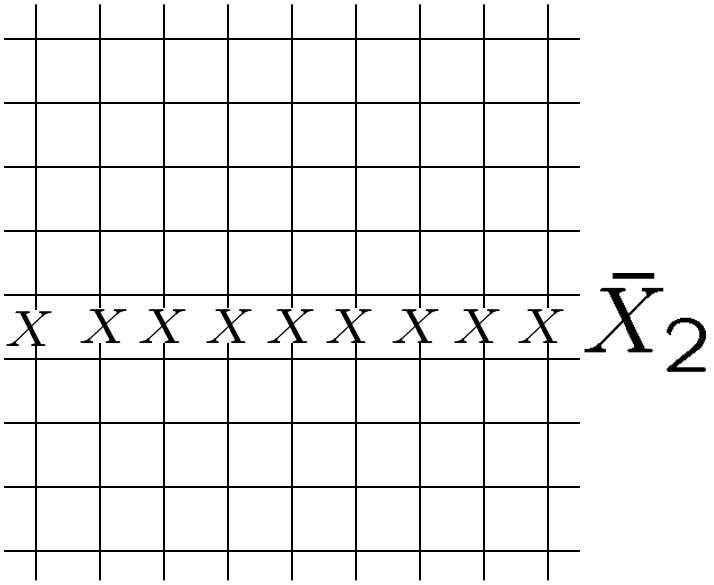


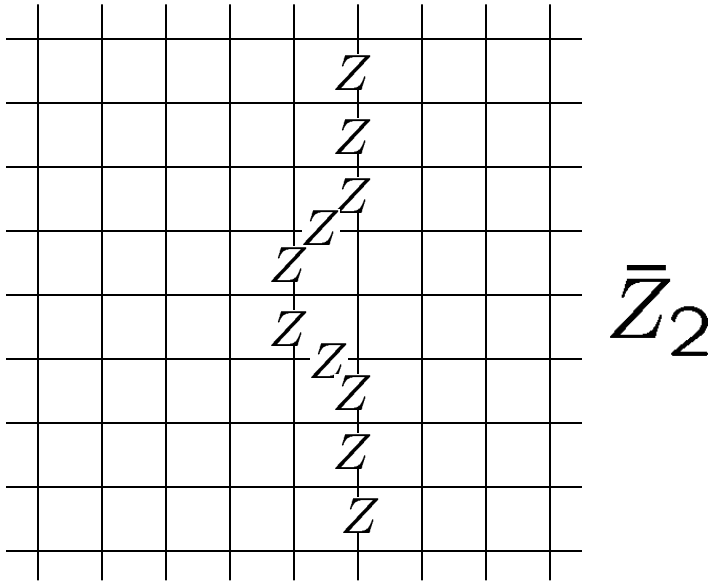
Moving an X or Z type particle on a homologically nontrivial path and annihilating with it's partner preserves the ground states

$$\bar{X}_1 \bar{Z}_1 \bar{X}_1 \bar{Z}_1 = -I$$

Action on ground state must be representation of non-Abelian group: Acts as encoded single qubit Pauli operator.

Two Qubits



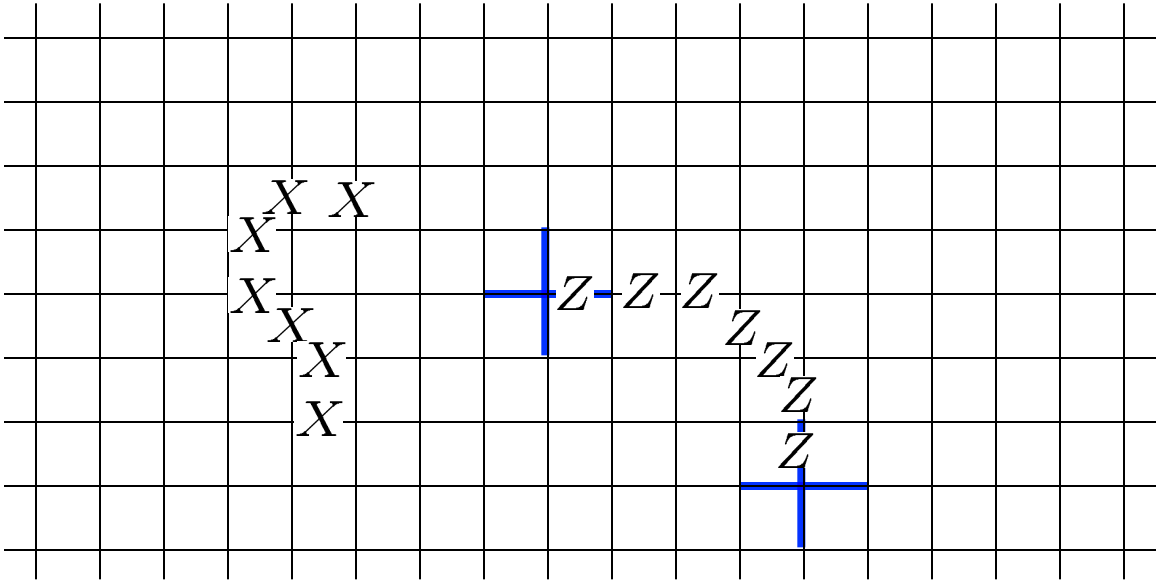


Moving an X or Z type particle on a homologically nontrivial path and annihilating with it's partner preserves the ground states

$$\bar{X}_2\bar{Z}_2\bar{X}_2\bar{Z}_2 = -I$$

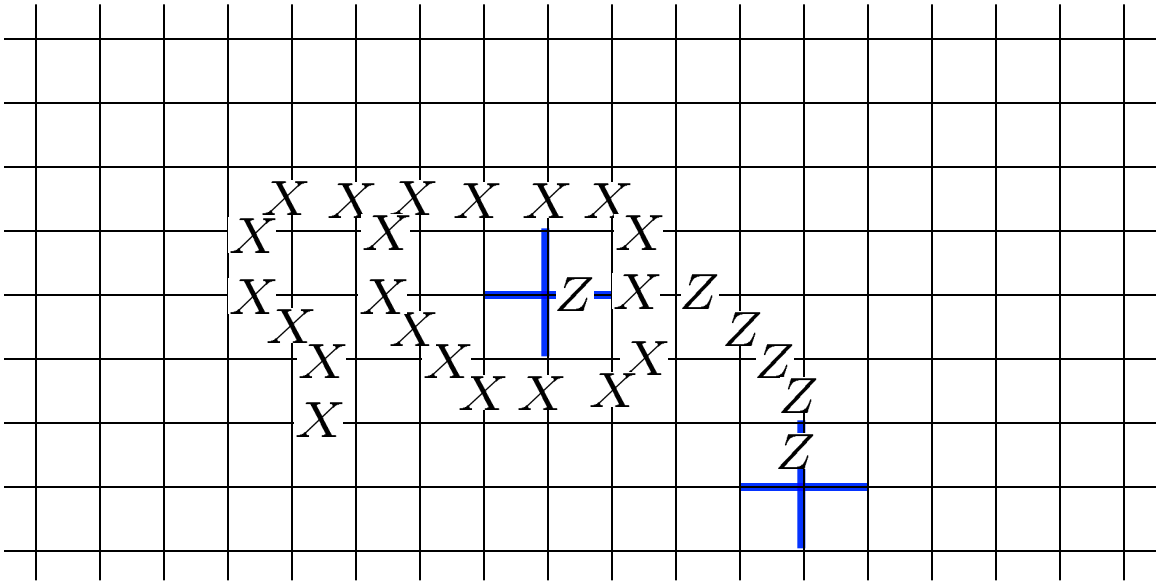
Action on ground state must be representation of non-Abelian group: Acts as encoded single qubit Pauli operator.

Anyons Anyone?



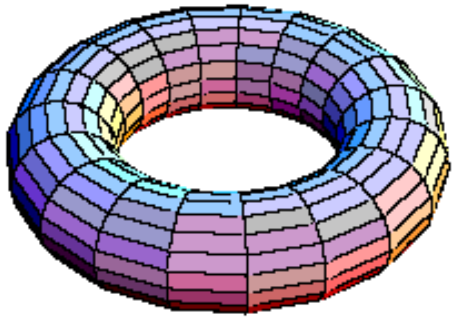
Phase:

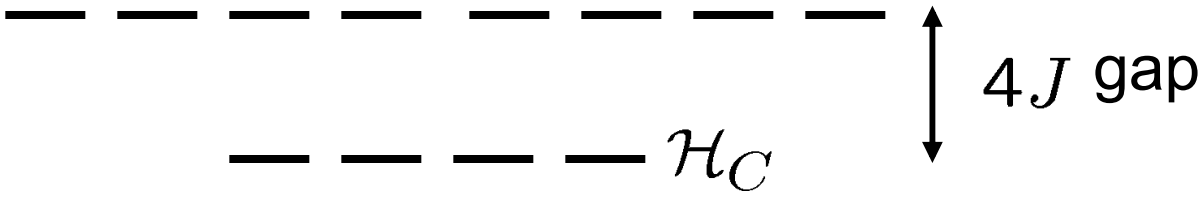
-1



Topological P[illegible]

Encode two qubits
into the ground state





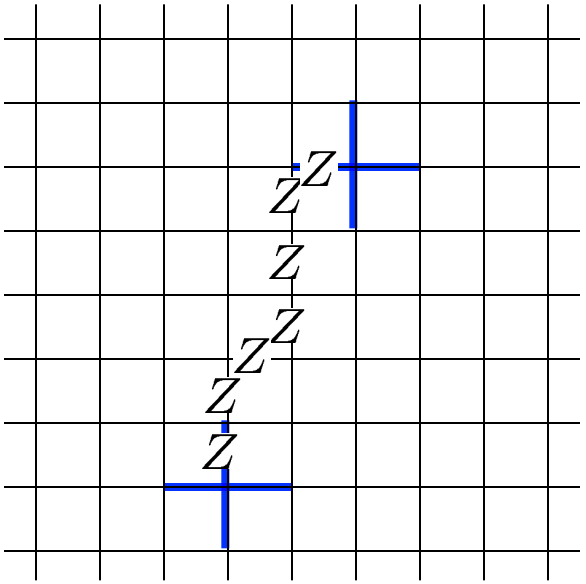
Perturbation theory:

$$V = \sum_v E_v + \sum_{vw} F_{vw}$$

But for $m \leq \left\lfloor \frac{l}{2} \right\rfloor$

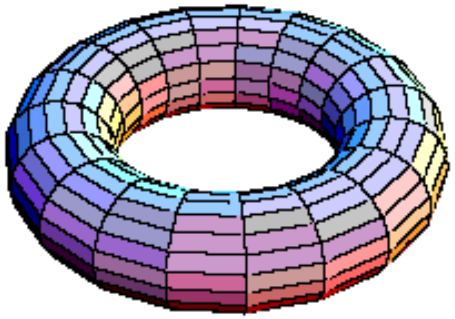
$$\langle \phi_i | V^m | \phi_j \rangle = 0$$

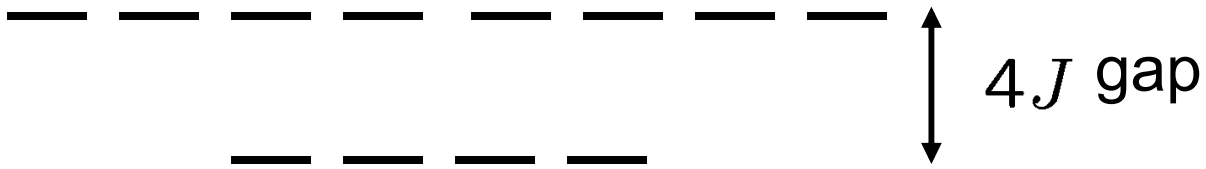
$$\langle \phi_i | V^m | \phi_i \rangle - \langle \phi_j | V^m | \phi_j \rangle = 0$$



Topological Quantum Computing

Encode two qubits
into the ground state





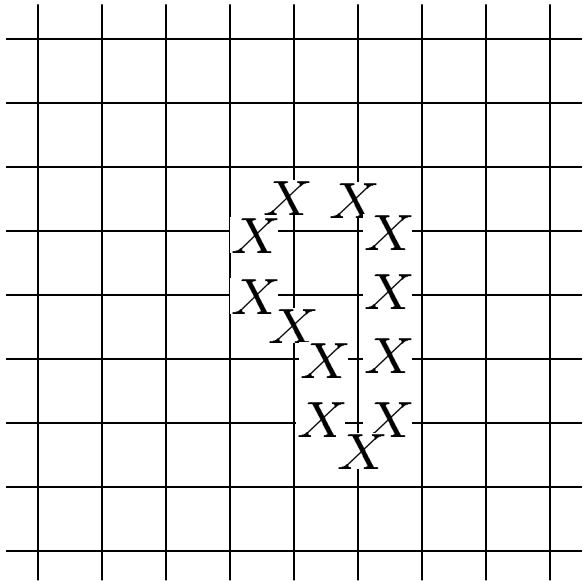
If we keep $kT \ll 4J$ then quantum information encoded into ground state will be “topologically” robust to perturbation.

We can perform gates (in this example X and Z) by creating anyon pairs and moving them around the torus.

Are these assumptions reasonable?

Topological Error Correction

Different mode of using Kitaev's Model, as a component in a fault-tolerant quantum computer, encode into ground state



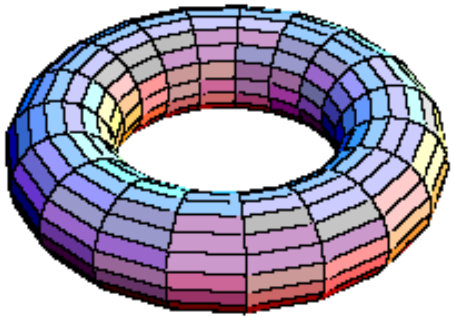
1. Errors occur
2. Measure vertex and plaque operators
3. Apply recovery operator
4. If created loop is homologically trivial, then we have restored information encoded into ground state.

Full analysis: measurements uncertain, recovery may fail, etc.
Threshold for fault-tolerant storage of quantum information:

[Dennis, Kitaev, Landahl, Preskill 2002]
[Wang, Harrington, Preskill 2003]

Fault-Tolerant?

Encode two qubits
into the ground state





If we keep $kT \ll 4J$ then quantum information encoded into ground state will be “topologically” robust to perturbation.

We can perform gates (in this example X and Z) by creating anyon pairs and moving them around the torus.

Are these assumptions reasonable?

What is a Computer?

Practical question (as opposed to philosophical)

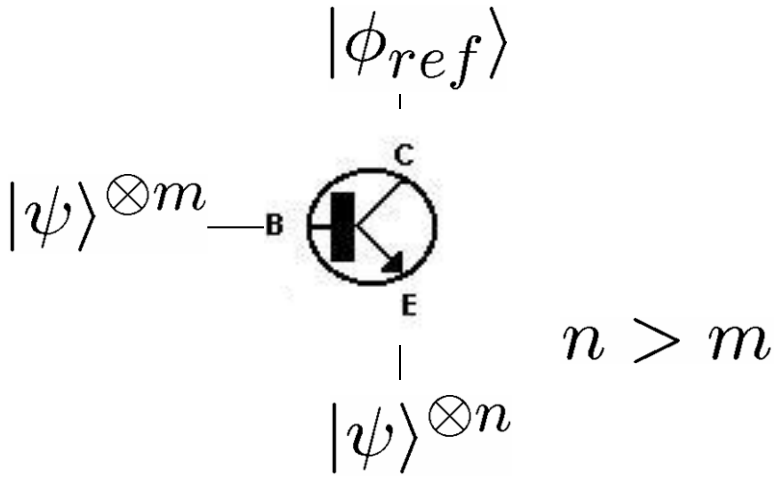
What is the phase of matter corresponding to the computer?

There are distinct physical reasons why robust classical computation is possible.

Hard Drive 

Integrated Circuit 

No Quantum Transistor?



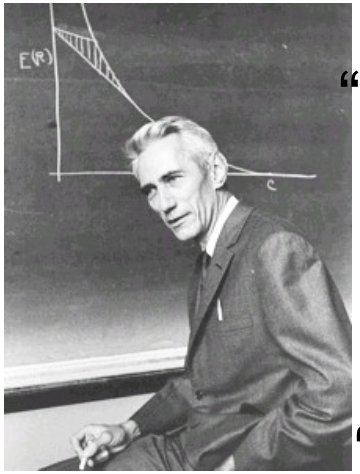


Quantum Cloning Machine

"A single quantum cannot be cloned," Wootters and Zurek, Nature, 1982

To Build a ~~Quantum~~ Computer
Classical

Problem (1938): How to build a reliable system from unreliable components?



Claude Shannon

"A mathematical theory of communication" (1948)

How to communicate in the presence of noise

"Probabilistic logics and the synthesis of reliable organisms from unreliable components" (1956)

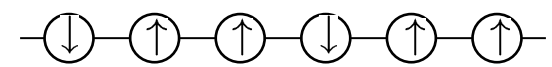


John von Neumann

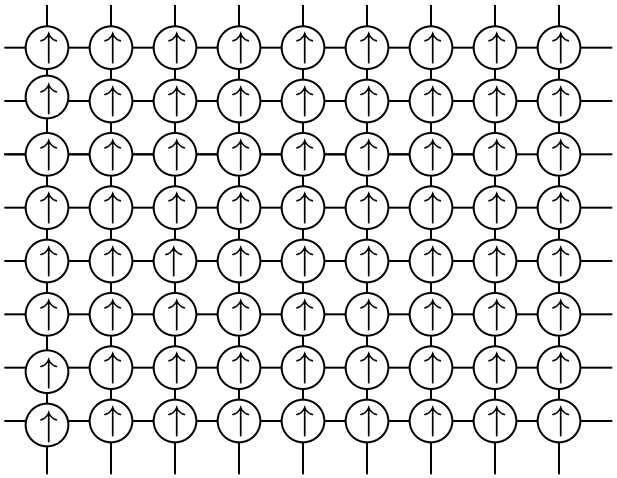
How to compute when the subsystems of your computer can fail?

Three Storage Testbeds

Ferromagnetic Ising models:



one dimensional



two (and >) dimensional

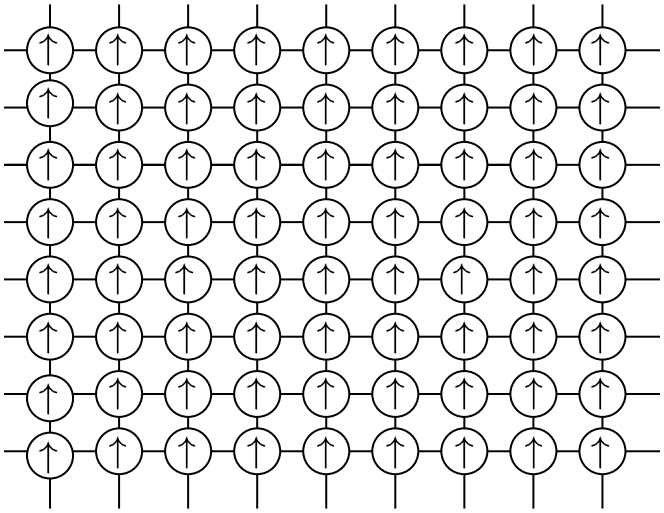
Toom's rule:

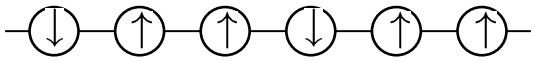
0	0	1	0	0	0	0	0
0	1	1	0	1	0	0	0
0	1	0	0	0	0	0	0
0	0	1	1	1	1	1	0
0	0	0	0	1	0	1	0
1	1	1	0	0	0	0	0

two dimensional noisy cellular automata

Ferromagnetic Ising Models







Ising model

spins on edges of a lattice: $s_i \in \{+1, -1\}$

energy of a configuration:

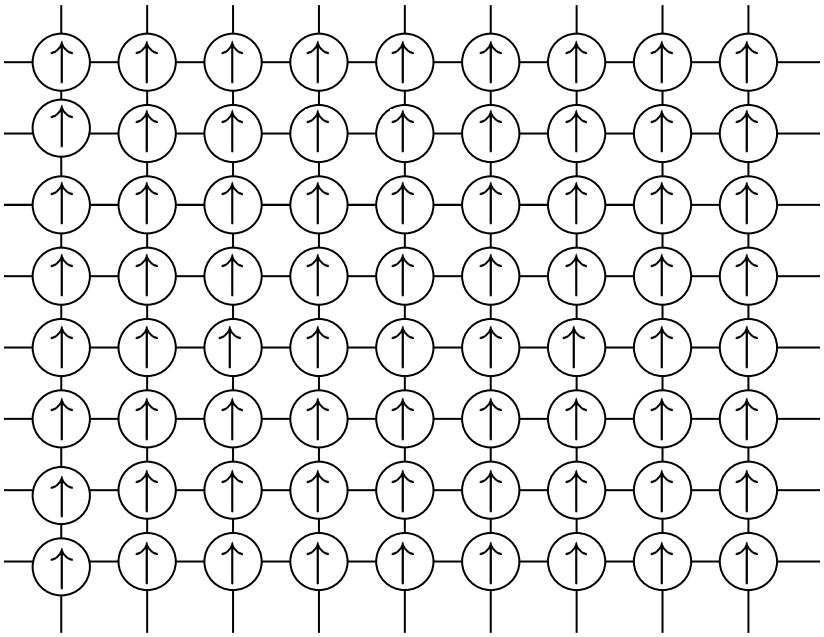
$$E = -J \sum_{\langle ij \rangle} s_i s_j + h \sum_i s_i \qquad J > 0$$

sum over neighbors

external magnetic field

$h = 0$ for now

Dynamic Ising Models



Dynamic Ising model (Metropolis update)

Interaction with environment causes spin flips

Update rule:

If flipping spin would decrease energy, flip spin

If flipping spin would increase energy, flip spin with probability

$\exp(-\beta \Delta E)$



Reminder



two-level system perturbatively coupled to a thermal boson bath

$$\frac{\partial \boldsymbol{\rho}}{\partial t} = -\bar{n}_{th}\frac{\gamma}{2}\left[\boldsymbol{\sigma}_-\boldsymbol{\sigma}_+\boldsymbol{\rho} + \boldsymbol{\rho}\boldsymbol{\sigma}_-\boldsymbol{\sigma}_+ - 2\boldsymbol{\sigma}_+\boldsymbol{\rho}\boldsymbol{\sigma}_-\right]$$

"heating"

$$-(\bar{n}_{th}+1)\frac{\gamma}{2}\left[\boldsymbol{\sigma}_+\boldsymbol{\sigma}_-\boldsymbol{\rho} + \boldsymbol{\rho}\boldsymbol{\sigma}_+\boldsymbol{\sigma}_- - 2\boldsymbol{\sigma}_-\boldsymbol{\rho}\boldsymbol{\sigma}_+\right]$$

"cooling"

$$-\phi\left[\boldsymbol{\rho} - \boldsymbol{\sigma}_z\boldsymbol{\rho}\boldsymbol{\sigma}_z\right]$$

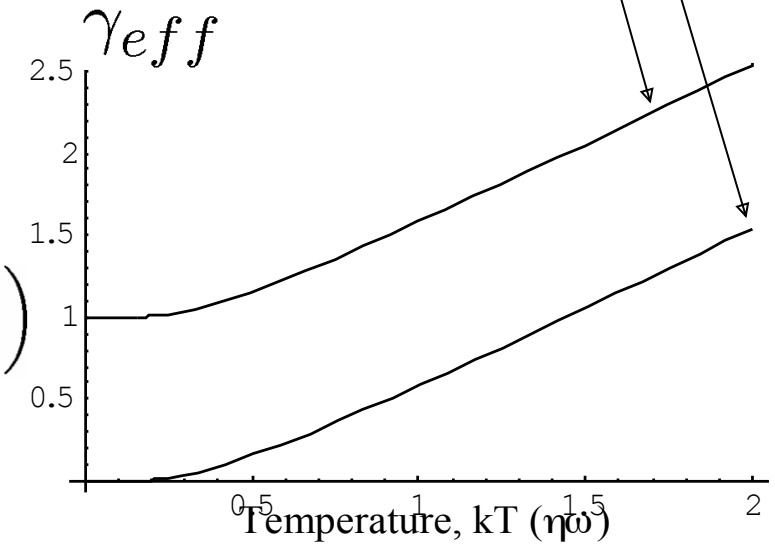
"non-dissipative"

$\hbar\omega$

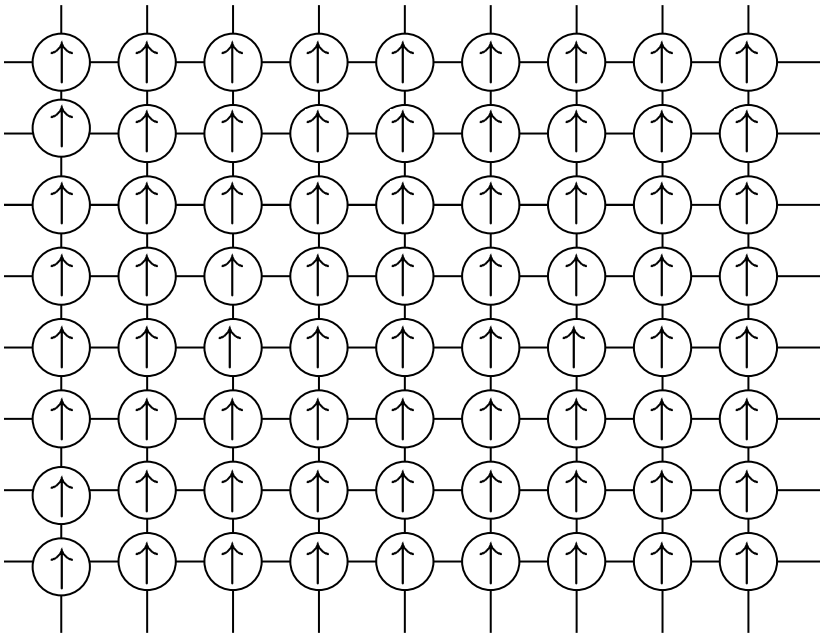
$$\bar{n}_{th} = \frac{1}{\exp\left(\frac{\hbar\omega}{kT}\right)+1}$$

$$\frac{\text{heating}}{\text{cooling}} = \frac{\bar{n}_{th}}{\bar{n}_{th}+1} = \exp\left(-\frac{\hbar\omega}{kT}\right)$$

$$= \exp\left(-\beta\Delta E\right)$$



Dynamic Ising Models



Dynamic Ising model (Metropolis update)

Interaction with environment causes spin flips

Update rule (alternate with noise):

If flipping spin would decrease energy, flip spin

If flipping spin would increase energy, flip spin with probability $\exp(-\beta \Delta E)$

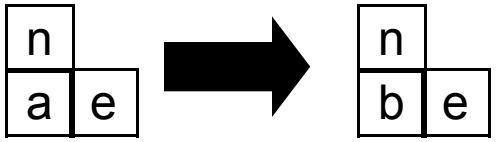
Toom's Rule



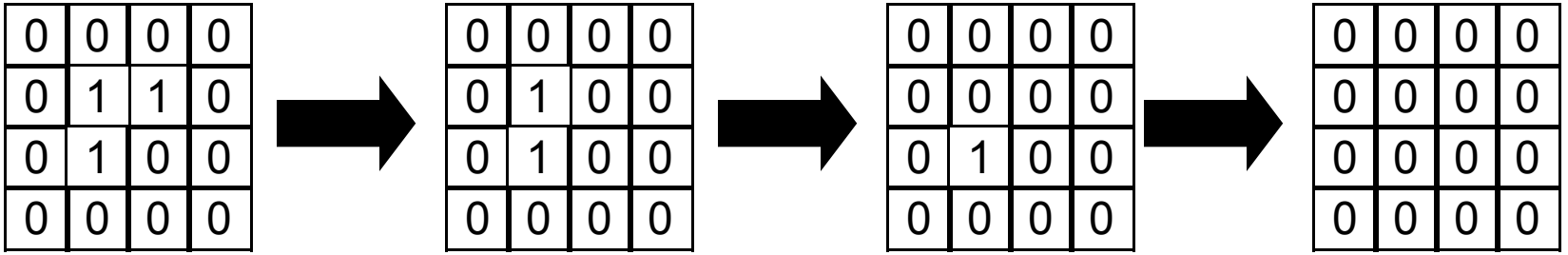
0	0	1	0	0	0	0	0
0	1	1	0	1	0	0	0
0	1	0	0	0	0	0	0
0	0	1	1	1	1	1	0
0	0	0	0	1	0	1	0
1	1	1	0	0	0	0	0

two dimensional noisy cellular automata

Update rule:



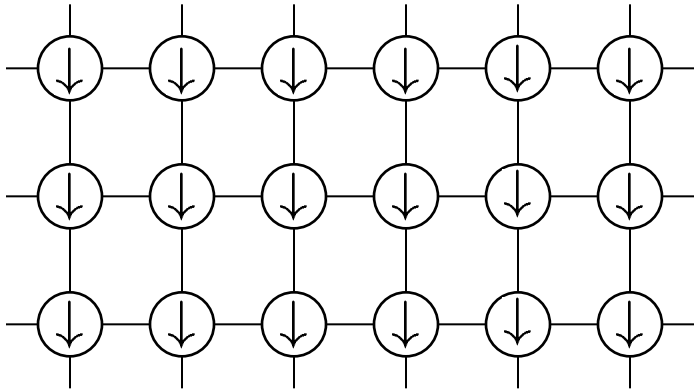
b=majority(n,e,b)



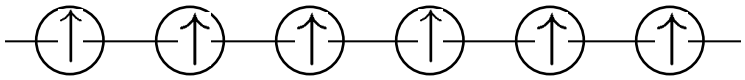
Update rule with noise: with some probability the update fails and a random state is assigned

Compare and Contrast

Encode information into the system and watch what happens



0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0



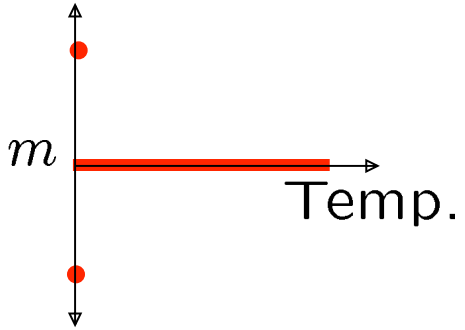
Redundancy code.

Order parameter is magnetization: $m = \frac{1}{N} \sum_i s_i$

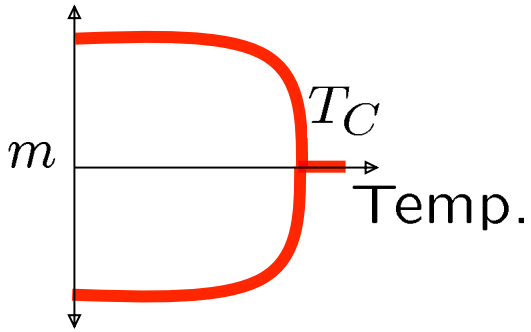
Storage

Long time behavior (not infinite time, but effectively)

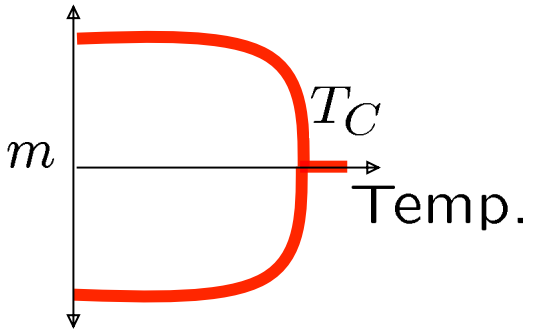
1D Ising



2D Ising



Toom's Rule



Criticism: (1) thermodynamic limit taken

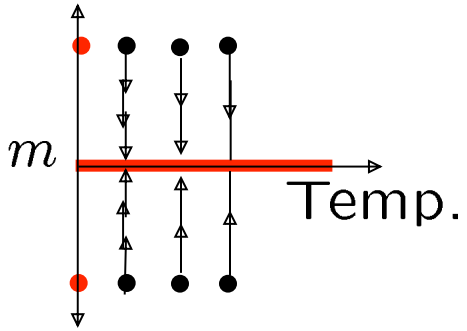
(2) what if relaxation to equilibrium takes a long time?

Storage



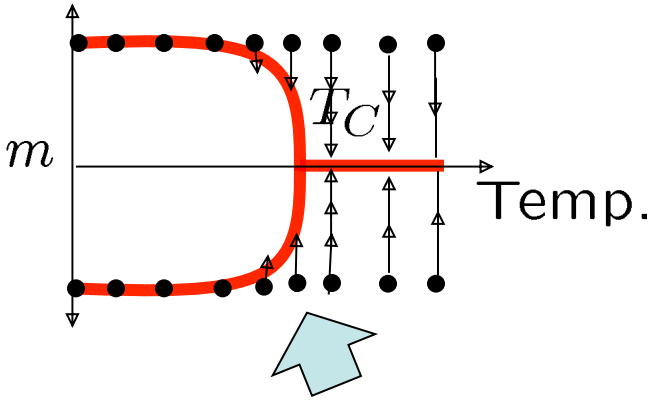
2nd attempt:
1D Ising

Relaxation to thermal



decay rate $\propto e^{-\beta J}$

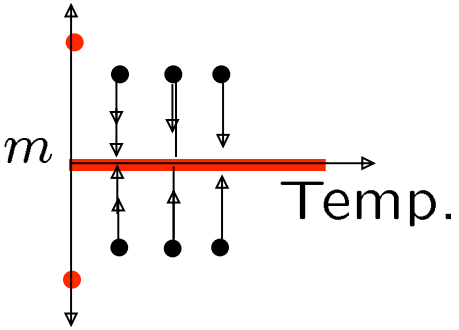
2D Ising, Toom's Rule

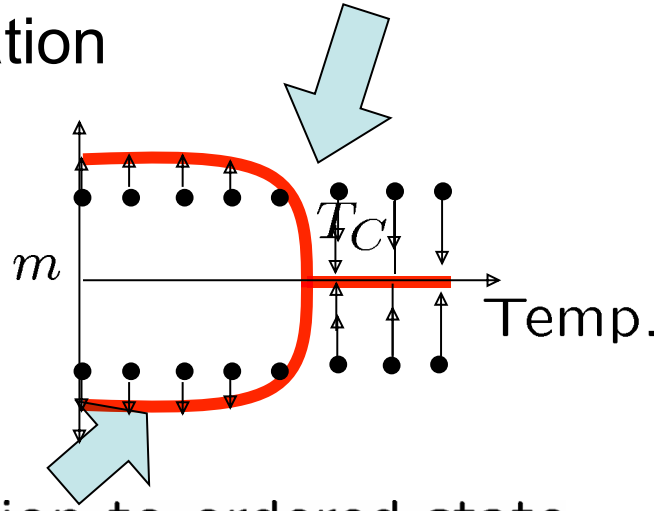


critical slowdown

$t_{th} \propto |T - T_c|^{z-\nu}$

Imperfect preparation





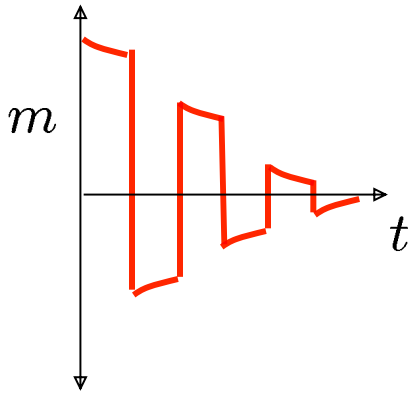
fast relaxation to ordered state

Manipulation



Flipping spins is imperfect

1D Ising

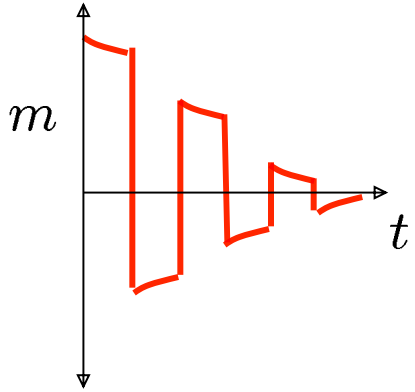


1D Ising
- temperature less than gap
- manipulation error must be small

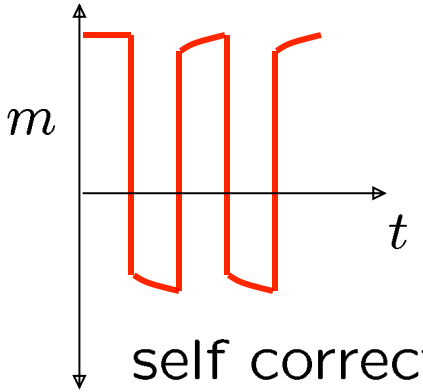
(i.e. NOT fault-tolerant)

2D Ising
- temperature less than critical temp

2D Ising
Toom's Rule



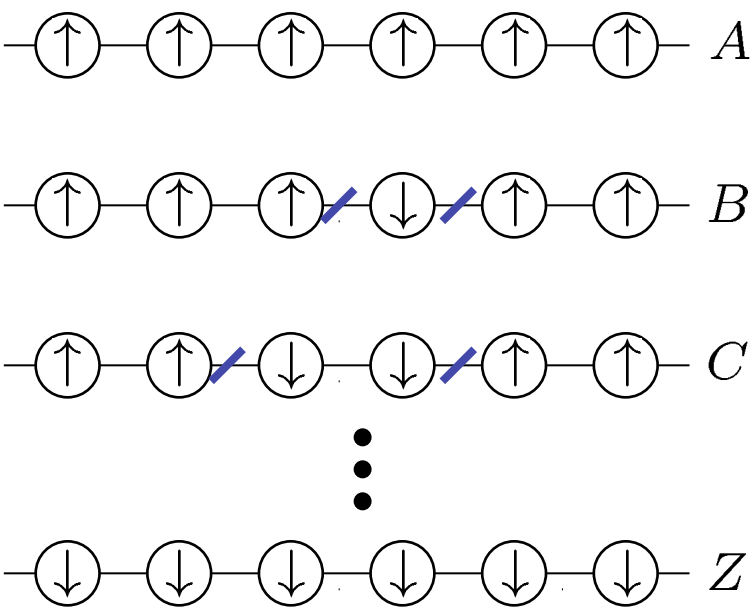
$T > T_C$

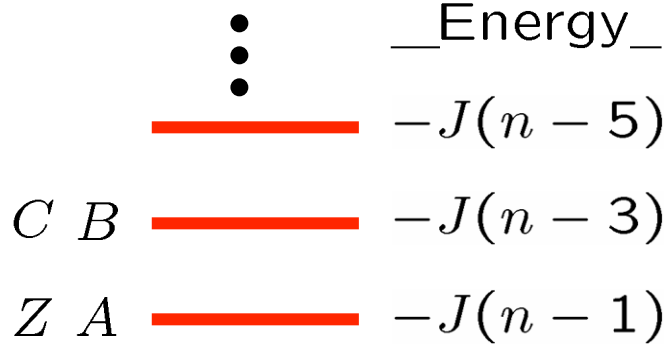


$T < T_C$

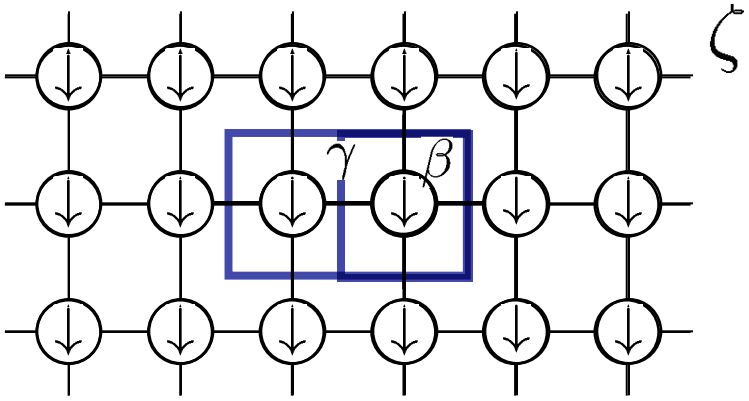
self correcting

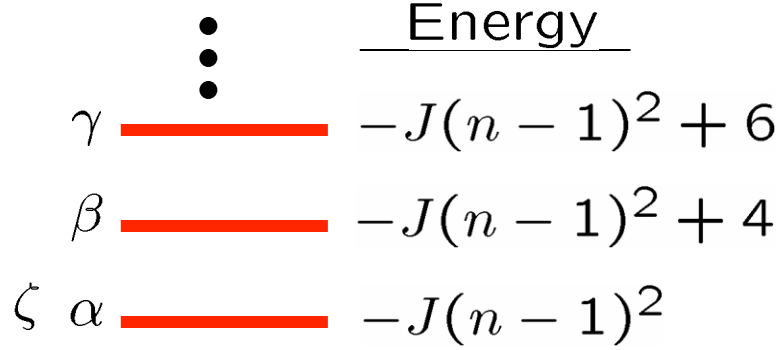
“Self Fixing”





Excitations point-like
Energy to err is constant
Disordering entropy >> Energy



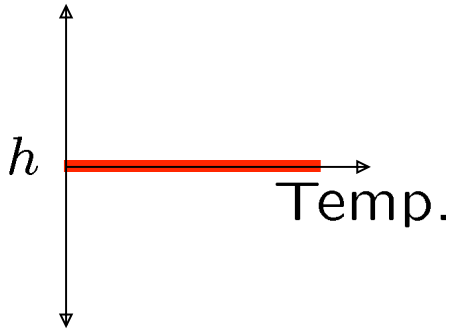


Excitations string-like
Energy to error is prop to size
Energy >> Disordering entropy

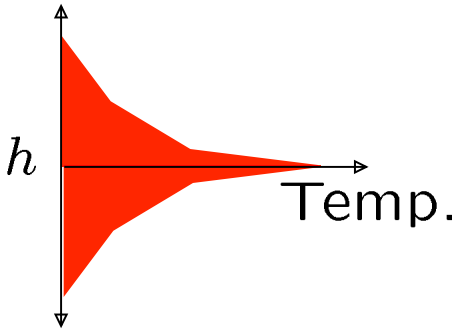
Toom's Rule Versus 2D Ising

Applied magnetic field (corresponds to a bias in noise)

2D Ising



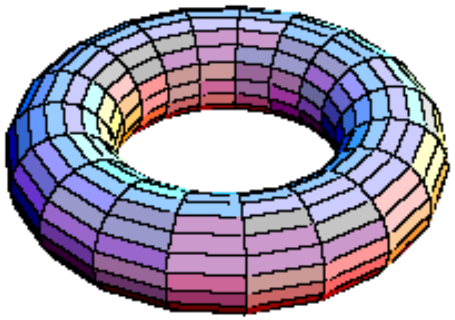
Toom's Rule



Regions with two coexisting phases

External magnetic fields disorder magnetic media

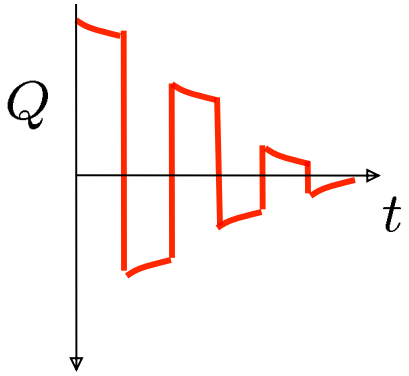
Whence Kitaev's Model?

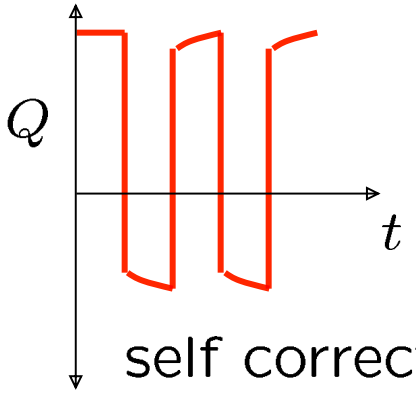


$$H = -J\left(\sum_p S_p + \sum_v S_v\right)$$

obust storage of quantum information.

ut it does not provide a mechanism for
bust manipulation of quantum information.





$T < T_C$

self correcting

Is it possible to construct a Hamiltonian, which allows for fault-tolerant manipulation as well?

Goal

1. Many-body quantum system with degeneracy due to existence of quantum error correcting code

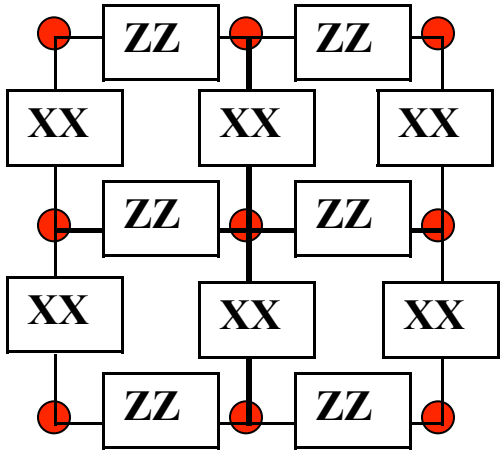
2. “Self-correcting”

3. Interaction which we could reasonably engineer

4. Ability to manipulate and measure erred quantum information.

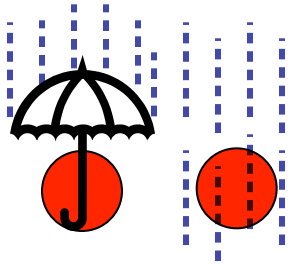
Quantum Compass Model

$$H = -J_X \sum_{ij} X_{i,j} X_{i+1,j} - J_Z \sum_{ij} Z_{i,j} Z_{i,j+1}$$



Ground state is a new kind of quantum error correcting code:
quantum error correcting subsystem

Subsystem Error Correction



Only need to protect subsystem, not full subspaces.

$$\mathcal{H}_{sys} = (\mathcal{C} \otimes \mathcal{D}) \oplus \mathcal{P}$$

encoding subsystem

$(|\phi\rangle\langle\phi| \otimes \rho_D) \oplus 0$ — $\mathcal{E}$ — $\mathcal{R}$ — $(|\phi\rangle\langle\phi| \otimes \tilde{\rho}_D) \oplus 0$

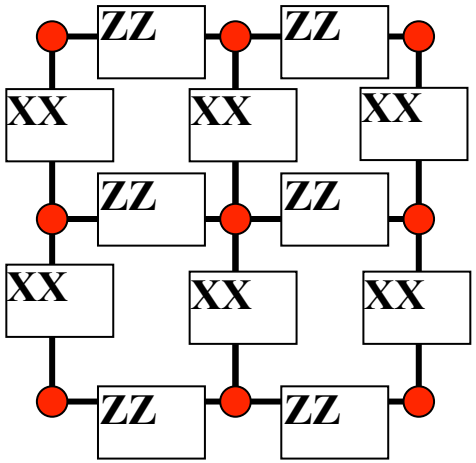
error recovery

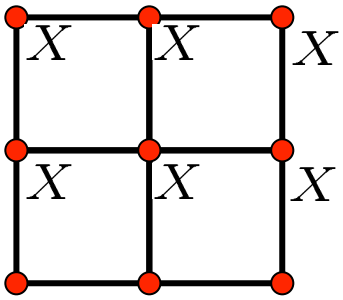
encoded quantum information

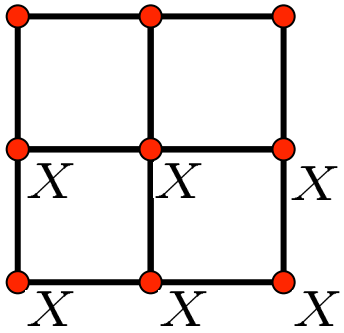
recovery need not perfectly restore system, but only preserve information encoded into subsystem.

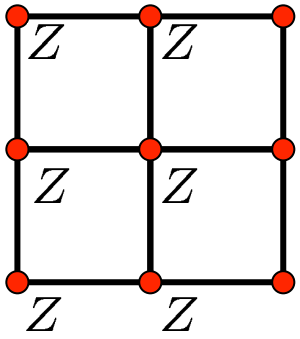
Quantum Compass Model

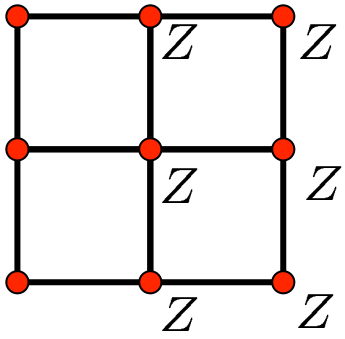
$$H = -J_X \sum_{ij} X_{i,j} X_{i+1,j} - J_Z \sum_{ij} Z_{i,j} Z_{i,j+1}$$

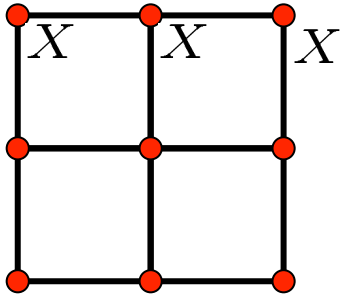


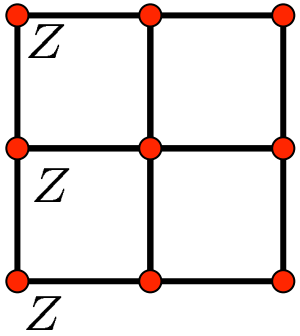






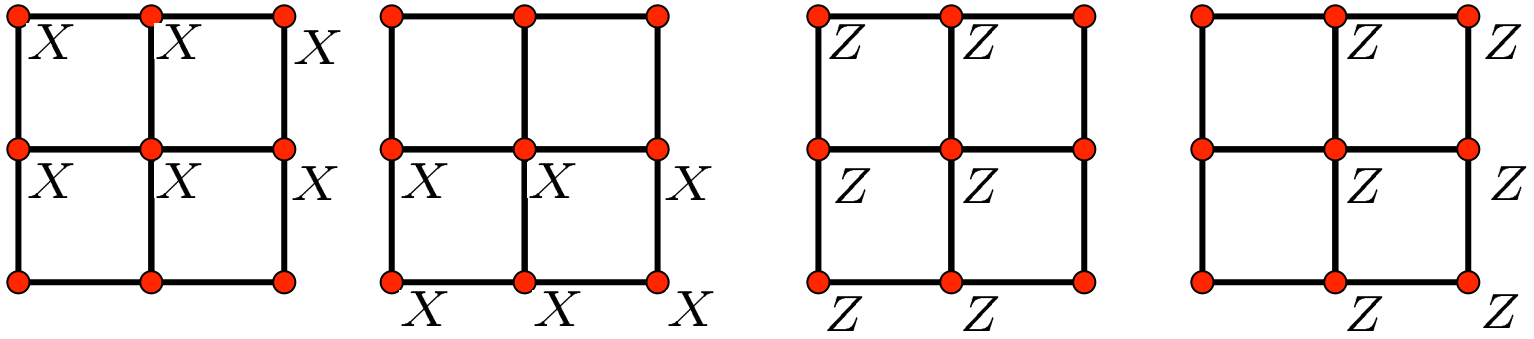




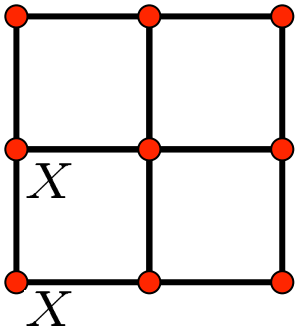


Subsystem Error Correction

Measure:

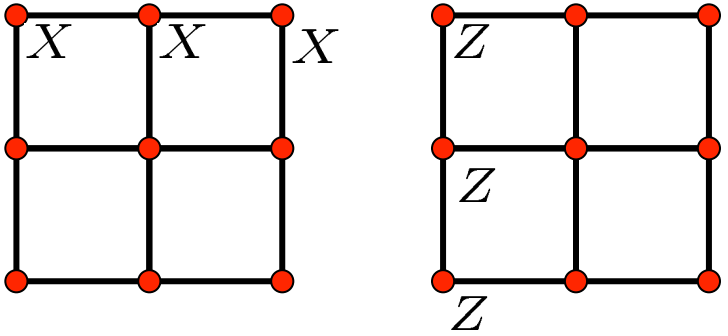


Diagnose error and apply fix. Fix may apply operator of form

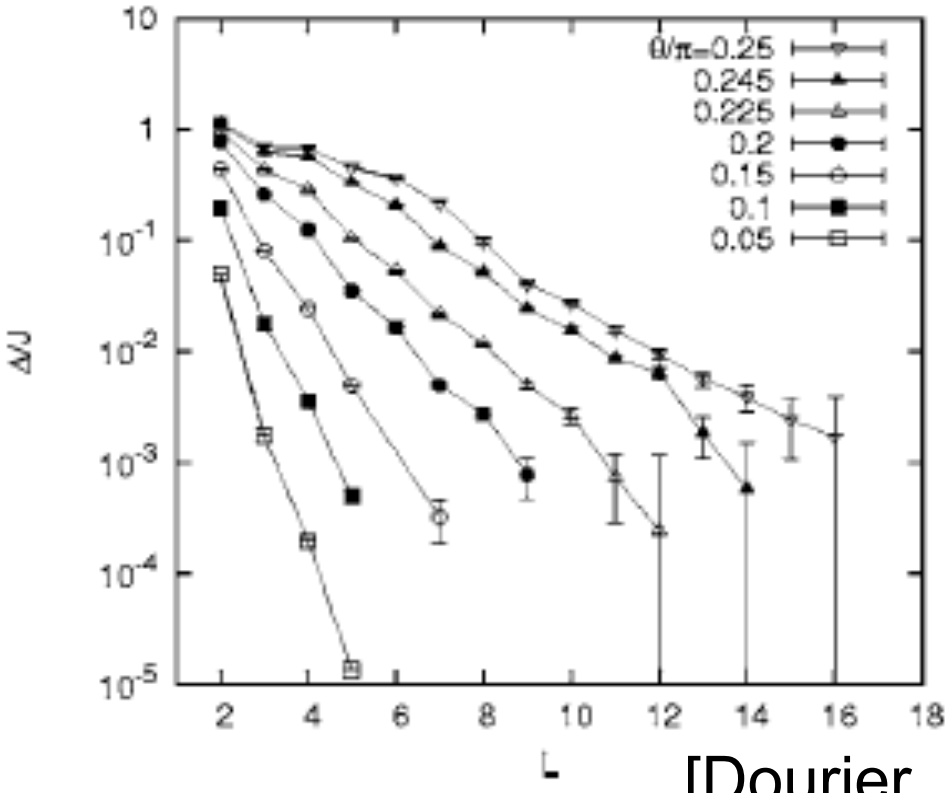


D. Bacon, Phys. Rev. A 73, 012340 (2006)
"Operator quantum error-correcting subsystems for self-correcting quantum memories"

But this commutes with ogical operators:



Problems:

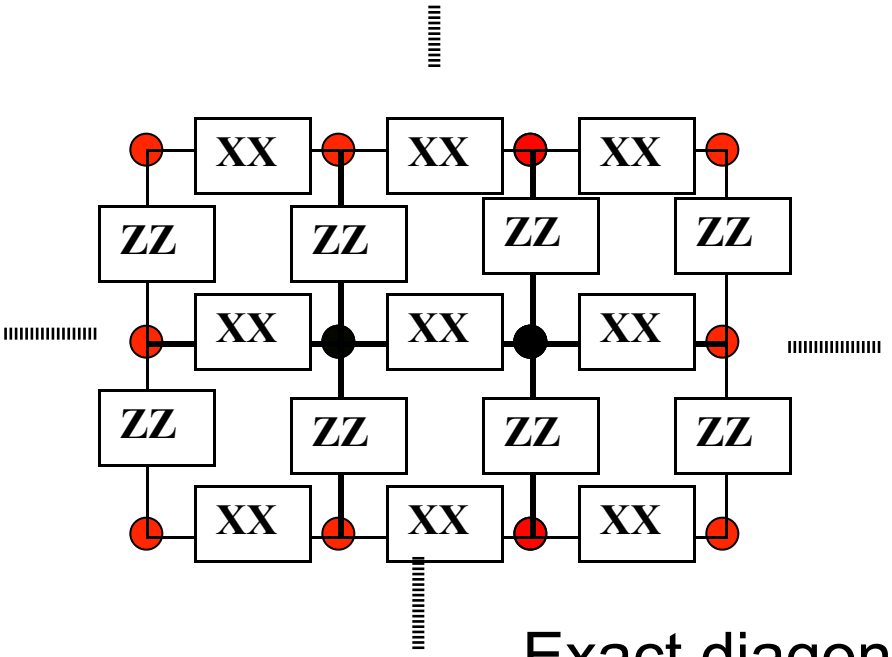


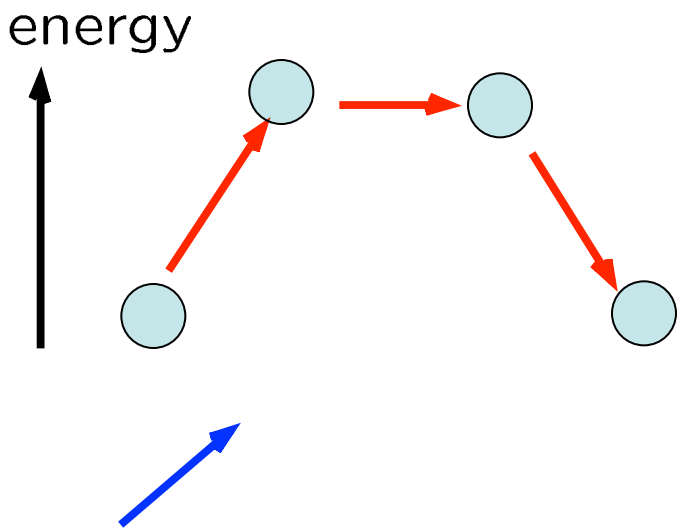
[Dourier, Becca, Mila 05]

$$H = -J \sum_{i,j=1}^{n} \cos(\theta) X_{i,j} X_{i+1,j} + \sin(\theta) Z_{i,j} Z_{i,j+1}$$

no gap in the thermodynamic limit

Problems, Continued



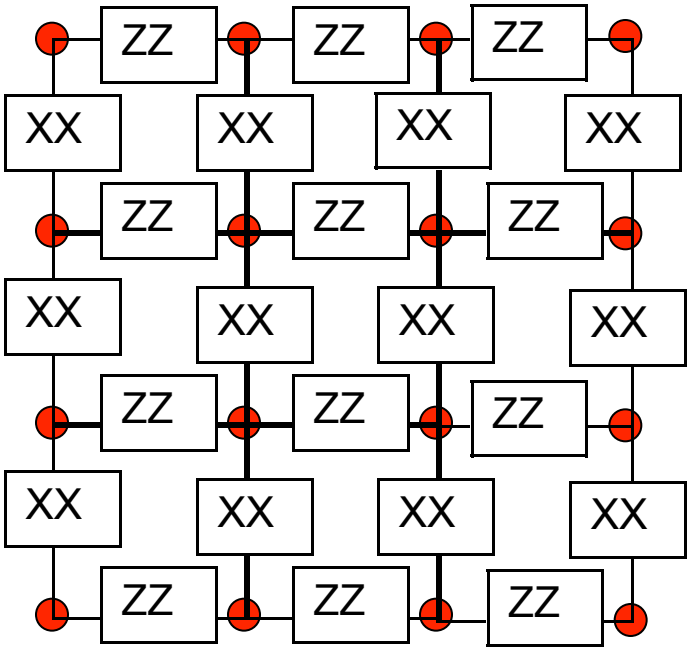


Exact diagonalization 3x3, 4x4, 4x5

Even at finite size, not “self-correcting”

Toward Self-Correction

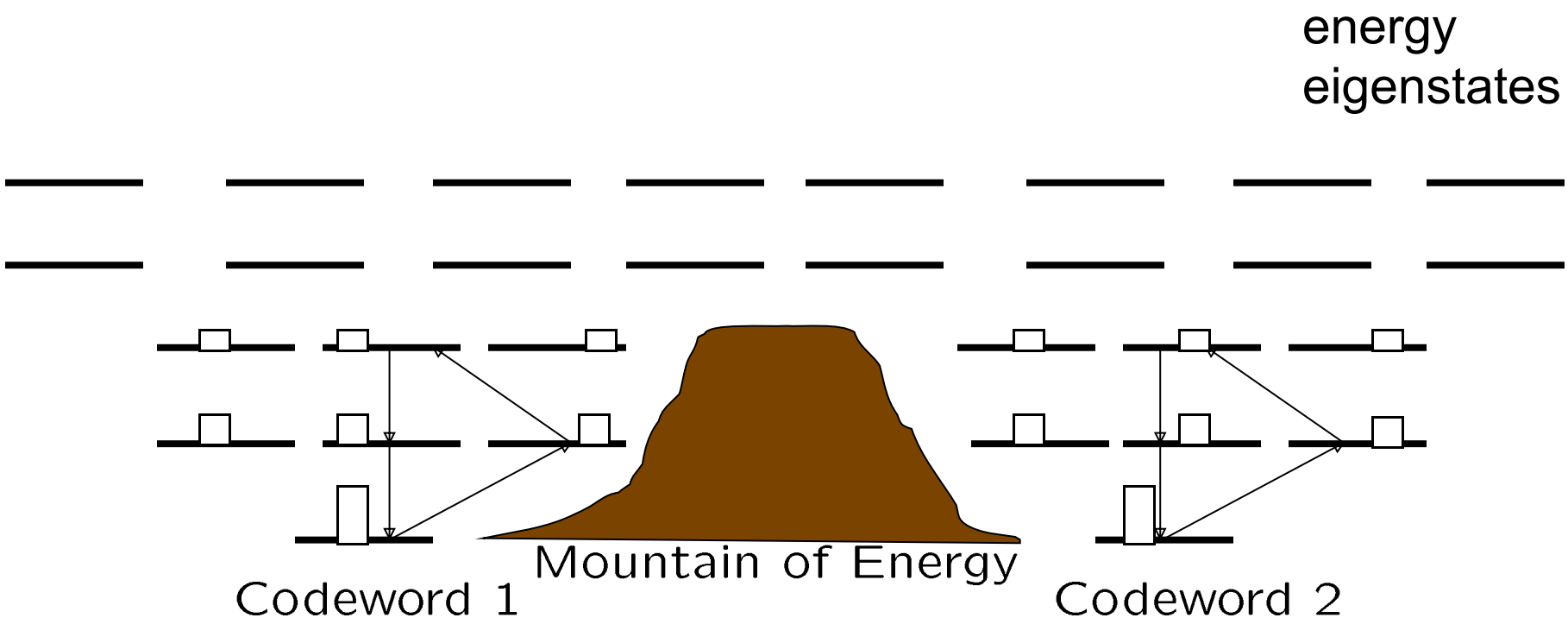
n by *n* square lattice



$$H = -\sum_{i,j=1}^{n}\sum_{k=1}^{n-1} J_X(k) X_{i,j} X_{i+k,j} + J_Z(k) Z_{i,j} Z_{i,j+k}$$

Exact diagonalization of 4x4 model, shows that with constant couplings, this model is self-correcting!

Self-Correction



Toward Self-Correction

Can techniques used in creating two qubit interactions be used to engineer such interactions
(Sorenson-Molmer gates? Ideas of Deng, Porras, and Cirac?)

Possible to engineer such systems in planar ion trap system?

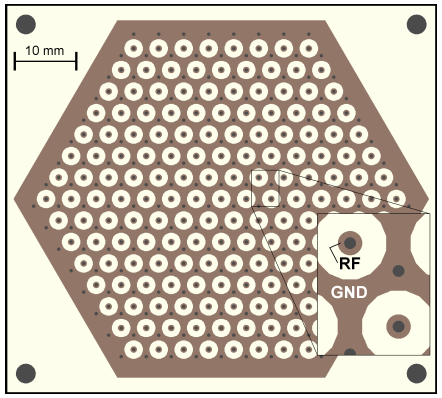
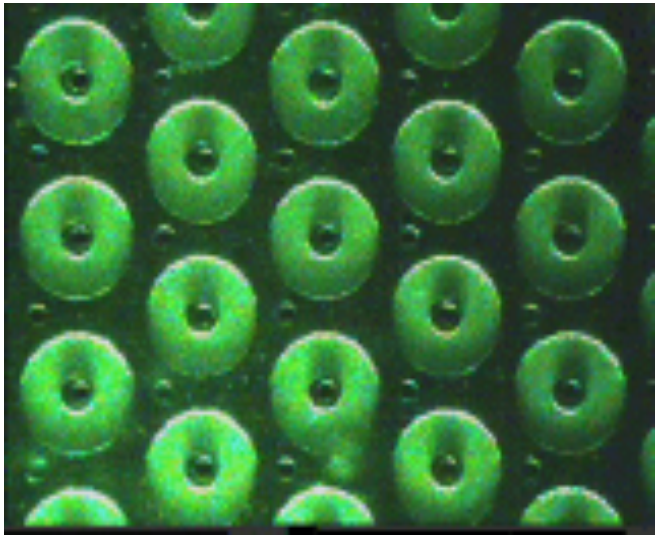


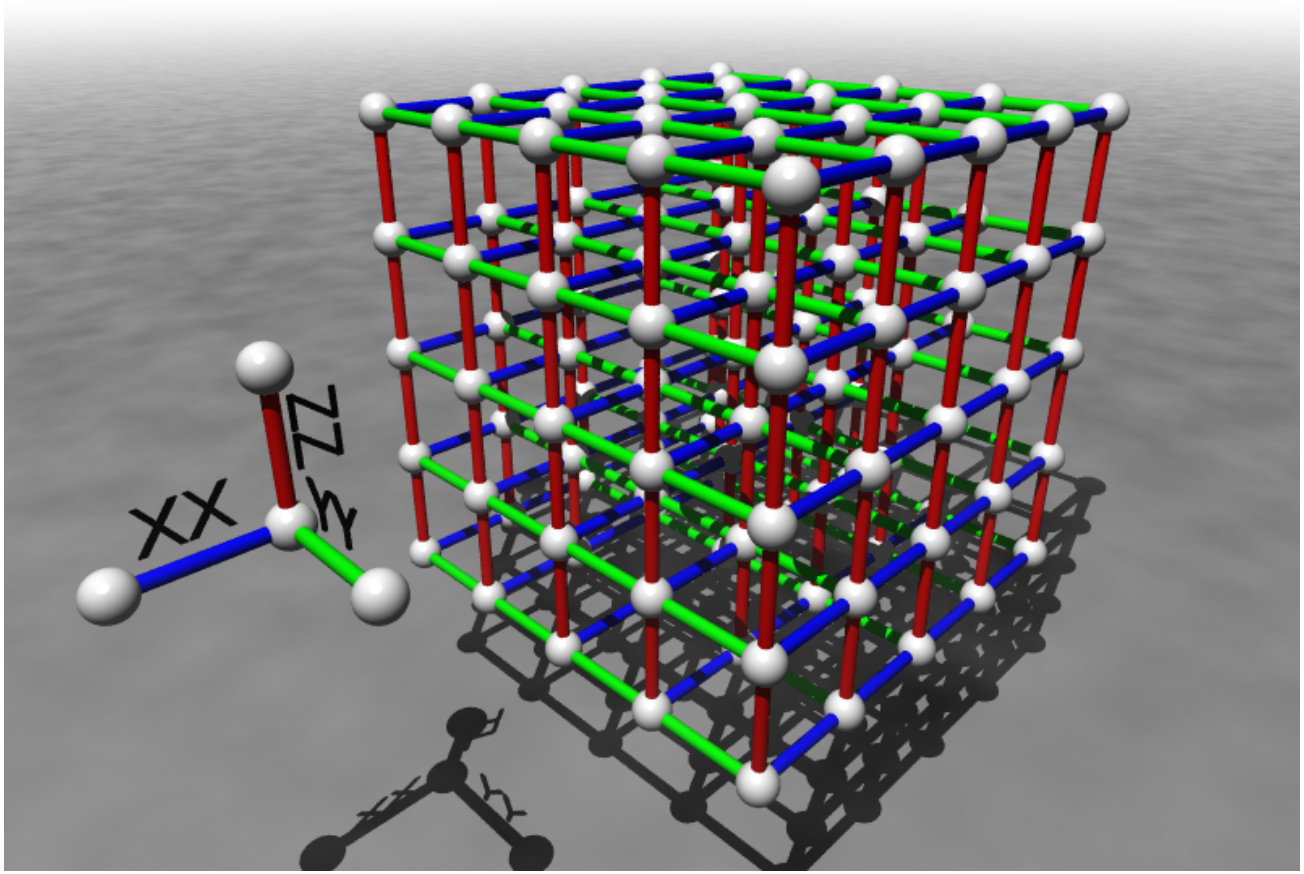
Figure 2. Hexagonal planar ion trap. 2D mult-zone ion traps allow one to trap ions in any spatial arrangement including a disordered lattice.



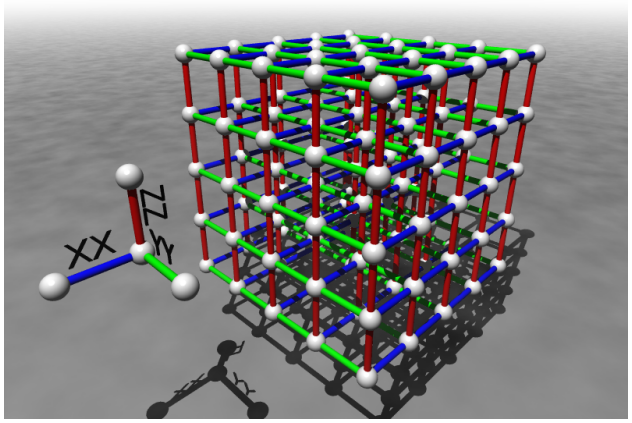
Picture courtesy Ken Brown
(Isaac Chuang's Lab, MIT)

Into The Third Dimension

$$H = -J_X \sum_{ijk} X_{i,j,k} X_{i+1,j,k} - J_Y \sum_{ijk} Y_{i,j,k} Y_{i,j+1,k} - J_Z \sum_{ijk} Z_{i,j,k} Z_{i,j,k+1}$$



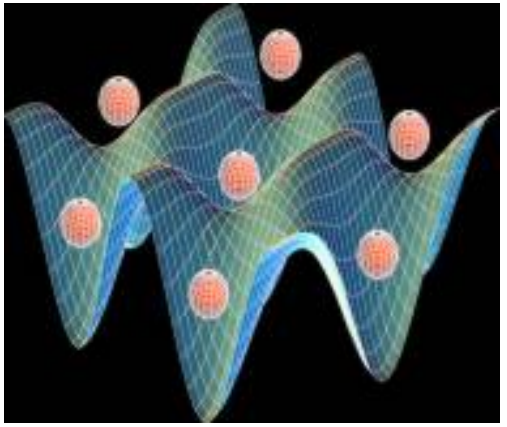
Into The Third Dimension



subsystem code

implementation in optical lattices?

Duan, Demler, and Lukin,
Phys. Rev. Lett. 91, 090402 (2003)



Current research: numerical work looking for self-correction [using new methods for density matrix renormalization group in greater than one dimension developed by Vidal, Verstraete et al.]

A Different Path



Quantum error correction can be performed by cooling an interacting quantum system.

Potential for robust substrates for quantum information.

Funding and Support

